2026 Wholesale Electricity Market Electricity Statement of Opportunities – Appendices

June 2026

A 10-year outlook of investment requirements to maintain reliability in the Wholesale Electricity Market in Western Australia's South West Interconnected System





We acknowledge the Traditional Custodians of the land, seas and waters across Australia. We honour the wisdom of Aboriginal and Torres Strait Islander Elders past and present and embrace future generations.

We acknowledge that, wherever we work, we do so on Aboriginal and Torres Strait Islander lands. We pay respect to the world's oldest continuing culture and First Nations peoples' deep and continuing connection to Country; and hope that our work can benefit both people and Country.

'Journey of unity: AEMO's Reconciliation Path' by Lani Balzan

AEMO is proud to have launched its Innovate Reconciliation Action Plan (RAP) in June 2026. 'Journey of unity: AEMO's Reconciliation Path' was created by Wiradjuri artist Lani Balzan to visually narrate our ongoing journey towards reconciliation – a collaborative endeavour that honours First Nations cultures, fosters mutual understanding, and paves the way for a brighter, more inclusive future.

Purpose

The Australian Energy Market Operator (AEMO) publishes the Wholesale Electricity Market (WEM) Electricity Statement of Opportunities (ESOO) under clause 4.5.11 of the Electricity System and Market Rules.

This publication is generally based on information available to AEMO as at 11 June 2026 unless otherwise indicated.

Disclaimer

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Modelling work performed as part of preparing this publication inherently requires assumptions about future behaviours and market interactions, which may result in forecasts that deviate from future conditions. There will be differences between estimated and actual results, because events and circumstances frequently do not occur as expected, and those differences may be material.

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Version control

Version	Release date	Changes
1	23/6/2026	Initial release
2	15/07/2026	Corrected two referencing errors. Added explanatory chart notes for Figure 10, Figure 11, Figure 13, and Figure 14.

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A1. Consumption and demand forecasting inputs

Each year, AEMO updates its 10-year projections of energy consumption and peak demand for the three Wholesale Electricity Market (WEM) scenarios and presents them in the WEM *Electricity Statement of Opportunities* (ESOO). These updates are based on stakeholder consultation and various model inputs.

For the 2026 WEM ESOO, AEMO aligned the *Low*, *Expected*, and *High* scenarios with the *Slower Growth*, *Step Change*, and *Accelerated Transition* scenarios respectively, which were developed under the Final 2025 *Inputs, Assumption and Scenarios Report* (IASR)[1] and subsequently revised under the Draft 2026 *Forecasting Assumptions Update*[2] in December 2025.

Each scenario considers alternative futures that meet Australia's net zero emissions commitment by 2050. The scenario mapping is broadly in line with the 2025 WEM ESOO as the scenarios still maintain relevance to reflect the changing landscape of social, technological, and political factors. However, a key revision compared to the 2025 WEM ESOO is the adoption of the *Accelerated Transition* scenario, which reflects slower hydrogen uptake and the exclusion of hydrogen exports across the forecast period, whereas in the 2025 WEM ESOO *Green Energy Exports* scenario faster hydrogen uptake and hydrogen exports was included.

AEMO provided external consultants with scenario assumptions to develop or update key inputs for the 2026 WEM ESOO consumption and demand forecasts. AEMO also consulted with stakeholders on the forecasting inputs via Forecasting Reference Group (FRG) meetings and 2025 IASR and *Forecasting Assumptions Update* consultations.

Key matters considered in the WEM ESOO forecasting are:

- **Input forecasts including:**
  - **Macroeconomic projections** – economic drivers at a national and state level, including economic growth, gross value added by sector, inflation, population, and dwelling construction,
    - state economic performance and demographic projections (Section A1.1), and
    - residential electricity connection forecasts (Section A1.2),
  - **Distributed photovoltaics (PV)** – including business and residential unscheduled installations (see Section A1.3).
  - **Electrification** – fuel-switching for industrial, commercial, and residential sectors (see Section A1.4**).**
  - **Electric vehicles (EVs)** – uptake of EVs (see Section A1.5).
  - **Distributed batteries** - uptake of distributed batteries (see Section A1.6).
- **AEMO consumer forecasts including:**
  - **Residential** forecasts (see Section A1.7).

[1] See: https://www.aemo.com.au/-/media/files/stakeholder_consultation/consultations/nem-consultations/2024/2025-iasr-scenarios/final-docs/2025-inputs-assumptions-and-scenarios-report.pdf?rev=63268acd3f044adb9f5f3a32b6880c27&sc_lang=en

[2] See: https://www.aemo.com.au/consultations/current-and-closed-consultations/draft-2026-forecasting-assumptions-update

- **Business mass market (BMM)** forecasts (see Section A1.8).
- **Large industrial load** forecasts in the South West Integrated System (SWIS – see Section A1.9), resulting in two forecasts:
  - existing large Industrial loads which are identified and categorised based on an extensive list of National Metering Identifiers (NMIs), and
  - new large Industrial loads which include an evaluative view of the pipeline of long-term projects informed by Western Power's assessment of project connections.
- **Energy efficiency forecast** (see Section A1.10) – state-based energy efficiency forecasts capture electricity savings arising from both autonomous efficiency improvements and policy-led energy savings across residential and business sectors.
- **New industries** forecasts (see Section A1.11) – hydrogen load assumptions and method for application to Peak Maximum and Minimum Demand forecasts.
- **Data centre** forecasts (see Section A1.12).
- **Distributed batteries** - impact on unscheduled operational demand (see Section A1.13).

These forecasting components served as key drivers for the consumption and demand forecasts. Among these drivers, AEMO recognised a markedly softened outlook for green hydrogen production based on recent market developments. This is reflected in reduced energy consumption for hydrogen production in the *High* scenario for the 10-year outlook period.

The 2026 WEM ESOO forecasts of distributed energy resources (DER) reflect:

- a continued growth of distributed PV,
- a stronger uptake of distributed energy storage systems (DESS) and virtual power plant (VPP) participation which co-ordinate operation of DER, and
- rebased combined residential and business EV forecasts, which are higher for the *Expected* scenario than 2025, informed by the latest available data as of March 2026.

Impacts from distributed PV[3], DESS and EVs are incorporated into the residential demand modelling process.

The Draft 2026 *Forecasting Assumptions Update*, published in December 2025[4], incorporates updated DER forecasts and revised generation technology costs and technical parameters to reflect recent technology developments. The report complements the broader components reported in the 2025 IASR.

A1.1 Economic and population growth outlook

AEMO engaged Deloitte Access Economics (DAE) to provide economic forecasts for Australia and the States and Territories, including Western Australia. DAE delivered the economic forecasts in October 2024[5] and delivered two subsequent

[3] Distributed PV includes PV non-scheduled generators (PVNSG), representing unregistered large PV systems ranging from 100 kilowatts (kW) to 30 megawatts (MW).

[4] See https://www.aemo.com.au/consultations/current-and-closed-consultations/draft-2026-forecasting-assumptions-update.

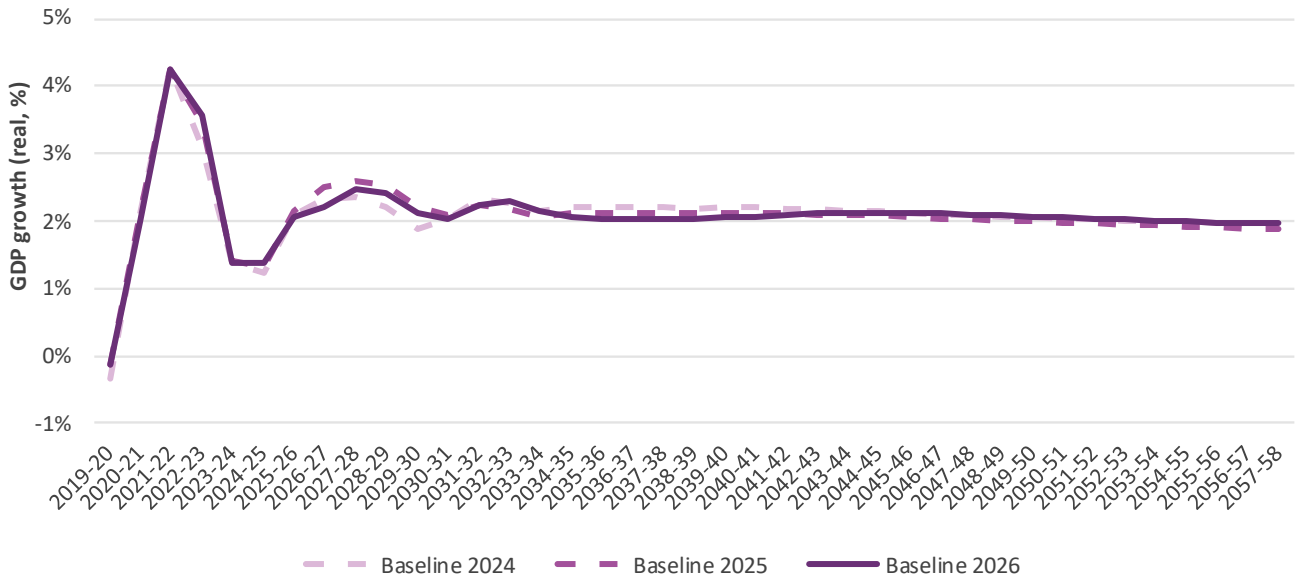
[5] See *Economic forecasts 2024-25,* 30 October 2024, Deloitte Access Economics, at https://www.aemo.com.au/-/media/files/major-publications/isp/2025/deloitte-access-economics-2024-economic-forecast.pdf?rev=92704425728b4097b3cb4894e1ccf0de&sc_lang=en.

updates in June 2025 and February 2026. The October 2024 economic forecasts were aligned to AEMO scenarios and provided several forecast series, including Gross Domestic Product, Gross State Product, Gross Value Added by industry, inflation, population, and dwelling construction. The two updates assessed evolving economic conditions but did not constitute full forecast revisions, and were provided as supplementary material for the 2025 and 2026 WEM ESOO. They included updates to DAE's Baseline economic projections, which are broadly aligned with AEMO's *Step Change* scenario. Both updates concluded that no significant long-term changes to the economic outlook had occurred since October 2024. Accordingly, AEMO continued to apply DAE's October 2024 forecasts as the basis for its economic modelling in the 2026 WEM ESOO.

Figure 1 presents real economic growth forecasts from DAE's internal 'Baseline' model for 2024 to 2026. While the plot indicates no material difference between the 2024 and 2026 Baseline series, the 2026 series reflects several changes in global and domestic economic conditions compared to the 2024 series, including:

- The trade policy of the United States has departed significantly from contemporary historical norms. The United States has levied and reversed tariffs at various rates to a multitude of countries, including Australia.
- Australia is exposed to the global economic and investment environment but is less directly exposed to changes in trade policy from the United States. As of December 2025, Australian goods exports to the United States only accounted for 5% of total goods exports.
- Global economic activity remained resilient despite elevated tariffs and increased geopolitical uncertainty. The global economy grew at a rate 3.3% in 2025, a rate broadly in-line with preceding years.
- Future downward revisions to Australian economic growth because of tariffs are minor, in part because of an assumption that Chinese policymakers implement stimulus measures to support the economy, in turn cushioning the Australian economy.
- Population and migration estimates for Western Australia have increased, despite decreasing at the national level.
- Productivity estimates and inflation estimates have been revised upwards.

Figure 1 Forecast real economic growth, 2019-20 to 2057-58, financial years (real, %)



Source: Deloitte Access Economics.
Note: presented on a financial year end basis.

AEMO is monitoring developments for potential impacts to economic activity following the commencement of the Middle East conflict on 28 February 2026. At this stage it is considered too early to provide a meaningful update.

A1.2 Residential electricity connection forecasts

AEMO's forecast of residential electricity consumption is driven by forecast growth in electricity connections. The number of electricity connections is a function of dwelling numbers, which are driven by population growth in Western Australia.

DAE provided updated dwelling forecasts as part of its early 2026 economic update. It forecasts a lower number of dwelling completions than previously anticipated, reflecting supply constraints in the construction industry that are limiting the pace of homebuilding activity. The slowest rate of construction is projected in the *Low* scenario, with higher rates in the *Expected* and *High* scenarios (see **Figure 2**). These differences reflect variations in the pace at which supply constraints are assumed to unwind, as well as differences in projected economic and population growth across scenarios.

AEMO used the above dwelling completion projections within a revised methodology to estimate residential connections, incorporating historical connection data provided by Synergy. The updated approach better accounts for dwelling demolitions, which in turn affects net connection growth rates. Improved treatment of demolitions has resulted in a downward revision to the connection forecast, as illustrated in **Figure 3**.

Figure 2 Actual and forecast Western Australian dwellings under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36

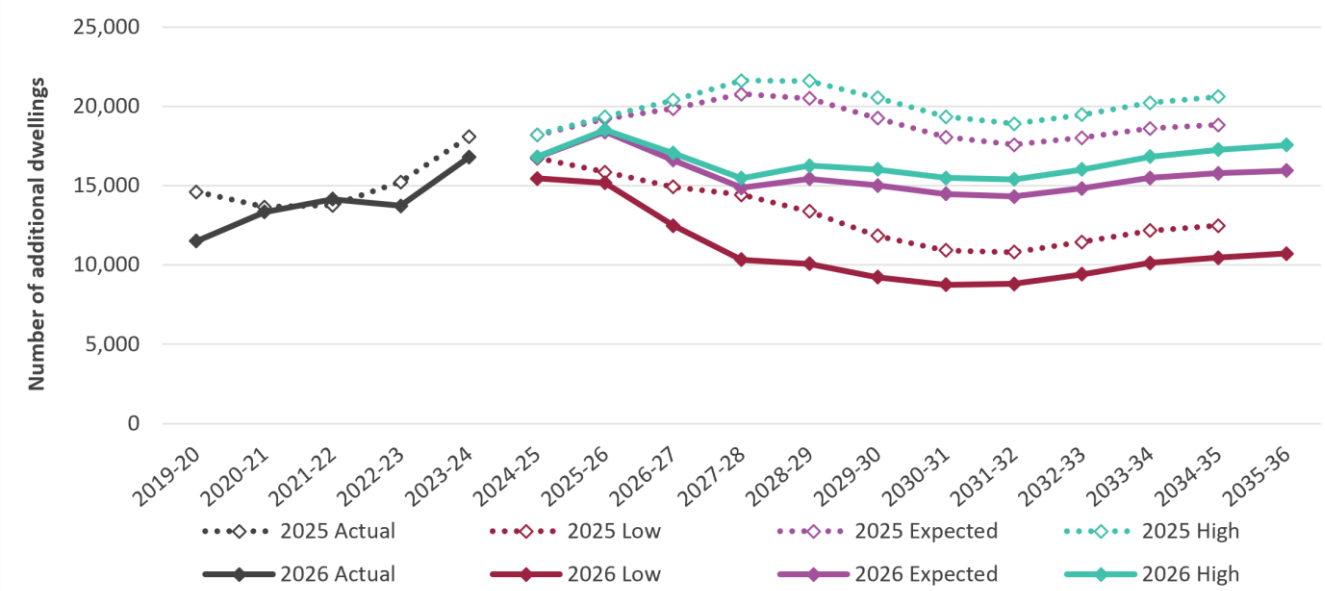
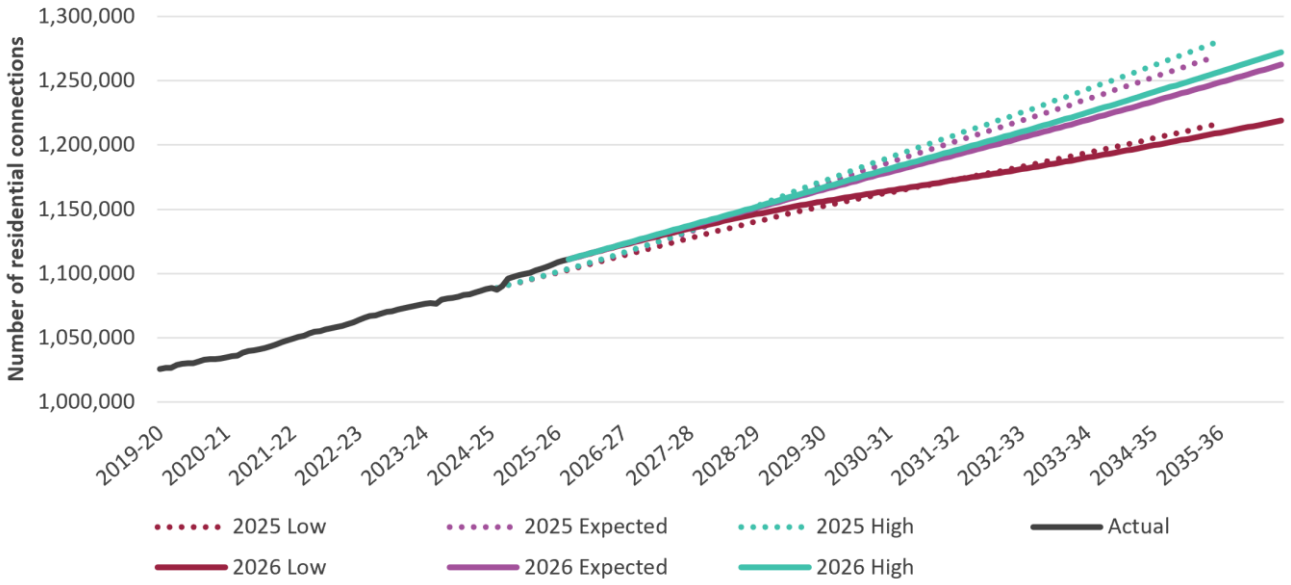


Figure 3 Actual and forecast SWIS connections under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36



A1.3 Forecast developments of distributed photovoltaics

Distributed PV installed capacity is expected to grow significantly over the outlook period across all three scenarios. This is driven by reductions in the cost of distributed PV systems, population growth, and the relatively short payback period for distributed PV investments. Installation of distributed PV capacity is supported to 2030 by the Small-scale Renewable Energy Scheme (SRES)[6], which provides Small-scale Technology Certificates (STCs) to consumers installing distributed PV systems. This is a subsidy provided under the national Renewable Energy Target.

As distributed PV costs have fallen, installed capacity has grown rapidly[7]. This trend is forecast to continue over the forecast horizon. Compared with last year, however, the scenario forecasts diverge more in the longer term. This reflects the forecasting assumptions update relative to the 2025 IASR, which shows increased uptake of distributed PV in the *High* scenario across the forecast period, increased uptake in the *Expected* scenario in the medium term, and a slightly reduced long-term uptake for the *Low* scenario. The long-term reduction in the *Low* scenario is largely the result of increased distributed PV costs relative to the prior assumptions. This is consistent with updated PV cost projections reflecting inflationary pressures in Commonwealth Scientific and Industrial Research Organisation (CSIRO) GenCost report[8]. The pace of distributed PV cost reductions is expected to slow. This is because solar panels now make up a smaller share of total distributed PV system costs, while installation costs, largely driven by Australian labour costs, are not expected to decline.

Figure 4 shows 2025 WEM ESOO and 2026 WEM ESOO residential distributed PV actuals and forecasts. Revised actuals have been included in the 2026 WEM ESOO forecasts to better account for unit replacements. **Figure 5** shows 2026 WEM ESOO actual and forecast installed business distributed PV capacity versus 2025 WEM ESOO.

[6] See https://cer.gov.au/schemes/renewable-energy-target/small-scale-renewable-energy-scheme.

[7] See Buckley and Dong (2023) *Solar pivot*: A massive global solar boom is disrupting energy markets and speeding the transition, Climate Energy Finance, June 2023.

[8] See https://www.csiro.au/en/research/technology-space/energy/Electricity-transition/GenCost.

Figure 4 Actual and forecast installed residential distributed photovoltaic capacity under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36 (megawatts [MW] degraded)

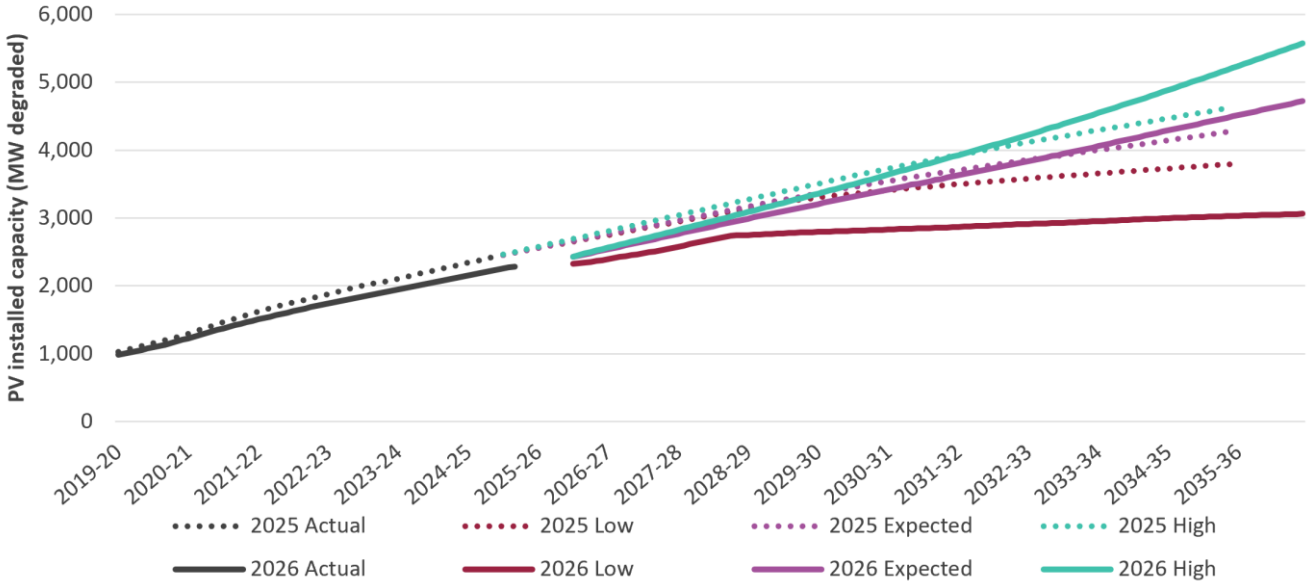
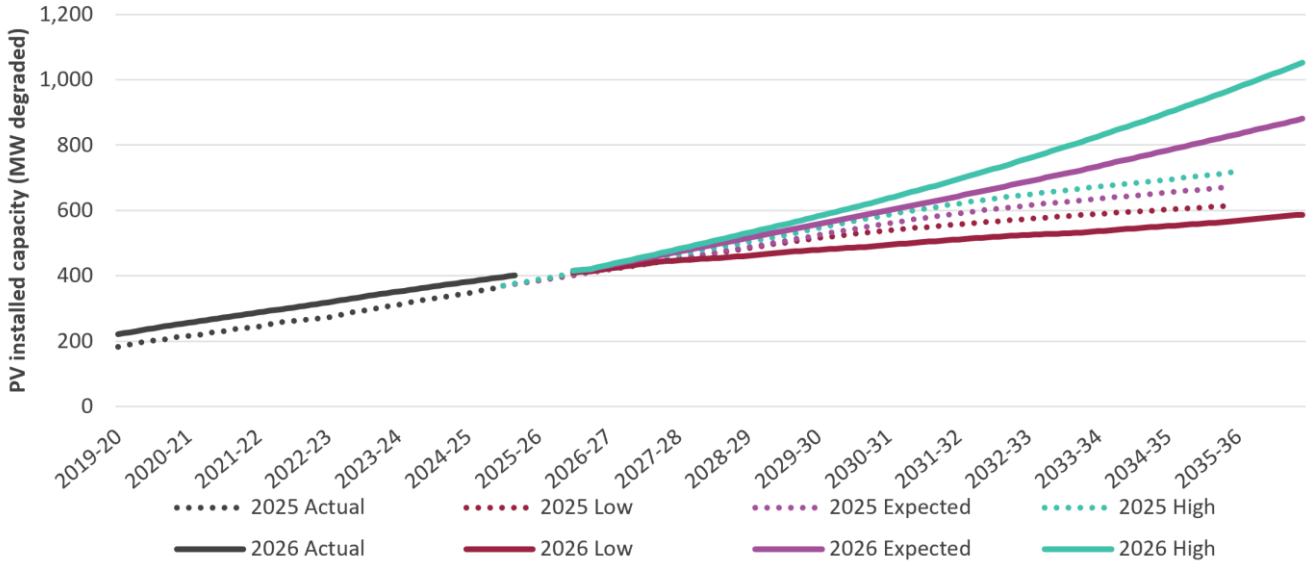


Figure 5 Actual and forecast installed business distributed photovoltaic capacity under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36 (MW degraded)



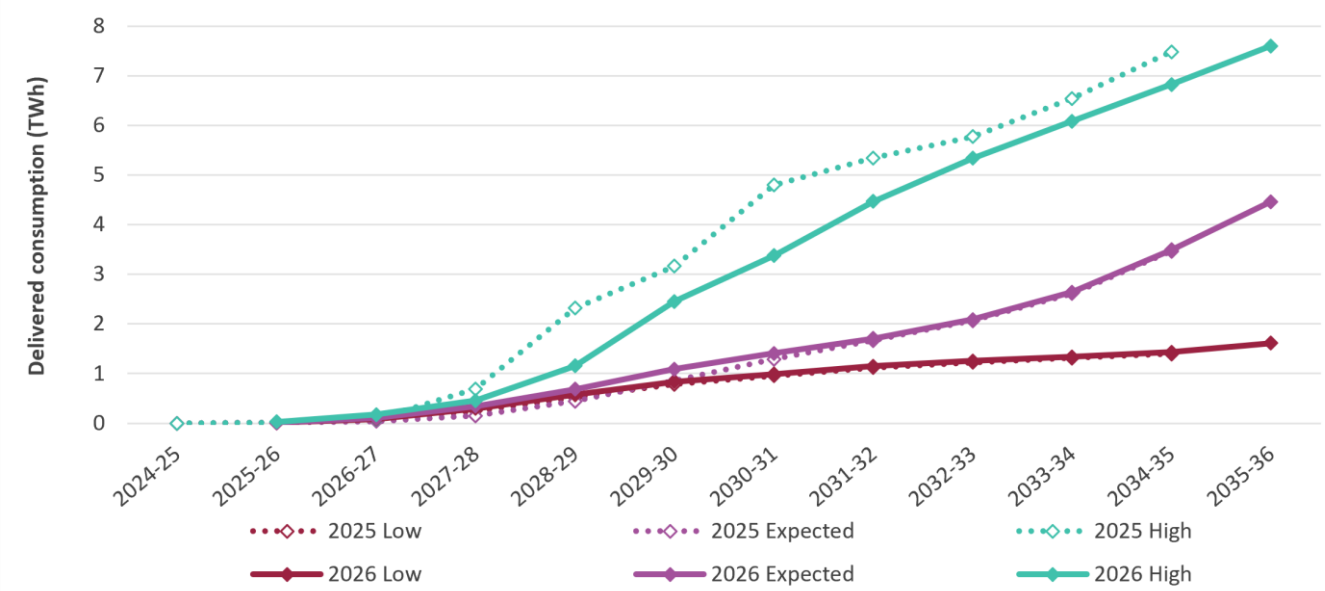
A1.4 Electrification

Ongoing growth in electrification is a key factor driving the increase in underlying electricity consumption across different sectors. AEMO's electrification forecasts included in this section cover the shift from gas to electricity in the residential sector (including space heating, water heating, and cooking) and in the business sector (such as industrial heating and mechanical processes). Transport electrification is addressed separately in the next section.

Figure 6 shows the electrification forecasts across the residential and business sectors in the SWIS. In all three scenarios, electrification is forecast to follow similar growth trends compared to the 2025 WEM ESOO. Growth is significantly greater

in the *High* scenario. By the end of the forecast period, additional electrification across the residential and business sectors is projected to account for approximately 18% of WEM grid-delivered consumption. Residential electrification is included within the residential consumption forecasts (see Section A1.7), and business electrification is presented separately (see **Figure 8** and **Figure 9** of Section 2.2.4 in the WEM ESOO).

Figure 6 Forecast delivered consumption resulting from electrification under three scenarios, from 2025 and 2026 WEM ESOOs, 2025-26 to 2035-36 (terawatt hours [TWh])



A1.5 Forecast development of electric vehicles

EV uptake in the SWIS is expected to remain broadly consistent with the 2025 WEM ESOO forecasts across all scenarios. Recent sales trends show EVs now account for around 12% of new vehicle sales[9].

Growth is being driven by both battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs), with scenario differences largely reflecting varying assumptions about the balance between hybrids and fully electric vehicles. The *Low* scenario assumes greater PHEV uptake and lower BEV uptake, while the *Expected* and *High* scenarios assume stronger BEV adoption as consumers increasingly seek to reduce emissions.

The forecasts were developed by CSIRO on behalf of AEMO and incorporate vehicle sales data to September 2025, EV charging and consumption profiles, and updated vehicle fuel efficiency assumptions. The modelling also reflects the impacts of national and state policies, including the New Vehicle Efficiency Standard (NVES)[10], Fringe Benefits Tax exemptions[11], and Western Australian Government EV initiatives. At this stage AEMO considers it too early to incorporate any meaningful impact on EV uptake from the commencement of the Middle East conflict on 28 February 2026.

[9] See https://www.aemo.com.au/-/media/files/stakeholder_consultation/consultations/nem-consultations/2025/draft-2026-fau/draft-2026-forecasting-assumptions-update.pdf?rev=eb147bbbe8c147dcb8a64f7a8b94e329&sc_lang=en.

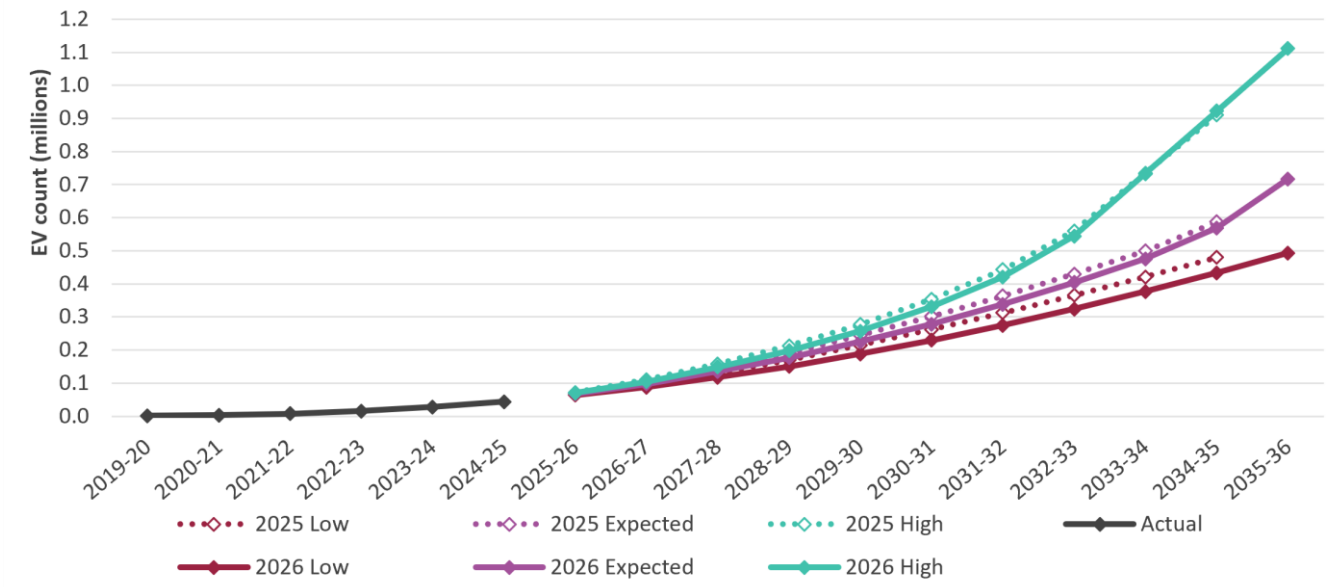
[10] See https://www.legislation.gov.au/C2024A00034/asmade/text.

[11] See https://www.ato.gov.au/businesses-and-organisations/hiring-and-paying-your-workers/fringe-benefits-tax/types-of-fringe-benefits/fbt-on-cars-other-vehicles-parking-and-tolls/electric-cars-exemption.

The Western Australian Government's EV initiatives combine financial incentives, infrastructure development, and a long-term plan for grid integration. Infrastructure efforts include the Western Australian EV Network, allocating $23 million for fast-charging stations along major routes, and the Charge Up grant scheme, covering up to 50% of EV charging infrastructure costs for local governments, small- to medium businesses, and not-for-profits. On public transport, $435 million in funding to electrify the Transperth bus fleet and install associated charging infrastructure and $6.2 million in installation of EV charging parking bays at Transperth train stations. The State EV Strategy for Western Australia supports the net zero emissions by 2050 by transitioning to EVs.

Figure 7 shows the forecast for BEV and PHEV fleet size in the WEM by scenario, compared to the 2025 WEM ESOO forecasts. The projected 2026 EV fleet size is similar to the 2025 forecast for all scenarios as the hybrids are considered as contributor to meeting the NVES targets in 2027-28. With the latest 2025 EV sales share projection, *Low* scenario assumes a greater penetration of hybrids and fewer EVs. While the *Expected* and *High* scenarios assume fewer hybrids and greater BEV uptake with the transport sector's electrification contributing more significantly to emissions reduction. The associated EV consumption is reflected in the respective residential (Section A1.7) and BMM (Section A1.8) forecasts.

Figure 7 Actual and forecast battery electric vehicle and plug-in hybrid electric vehicle fleet size under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36 (millions)



Note: presented on a Capacity Year end basis.
Source: CSIRO.

A1.6 Forecast developments of distributed batteries

All three scenarios of the 2026 WEM ESOO present a stronger uptake of distributed batteries than in last year's WEM ESOO, predominantly driven by the WA Residential Battery Scheme and the Commonwealth Cheaper Home Batteries Program[12].

Other drivers of the forecast uptake include projected technology cost reductions, the adoption of co-located batteries alongside distributed PV systems, and differences in average battery sizes across scenarios (smallest in the *Low* scenario, largest in the *High* scenario).

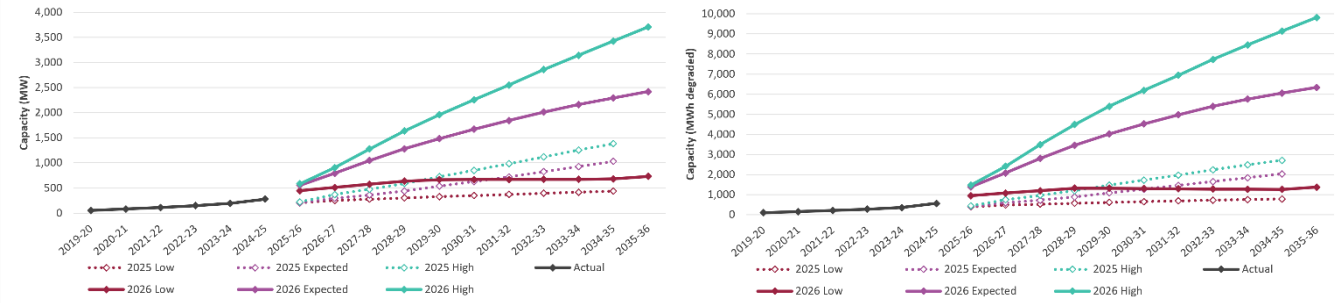
The forecast capacity uptake of distributed batteries is presented in **Figure 8**.

The WA Government's scheme is in addition to the Federal Government's Cheaper Home Batteries Program.

Under the WA Government scheme, battery applicants are eligible for a rebate or no-interest loan to purchase a home battery.

Batteries installed under the WA Residential Battery Scheme are required to participate in a virtual power plant (VPP) arrangement for at least two years.

Figure 8 Actual and forecast uptake of distributed batteries capacity under three scenarios from the 2025 and 2026 WEM ESOO, 2019-20 to 2035-36 (MW and megawatt hours [MWh] degraded)

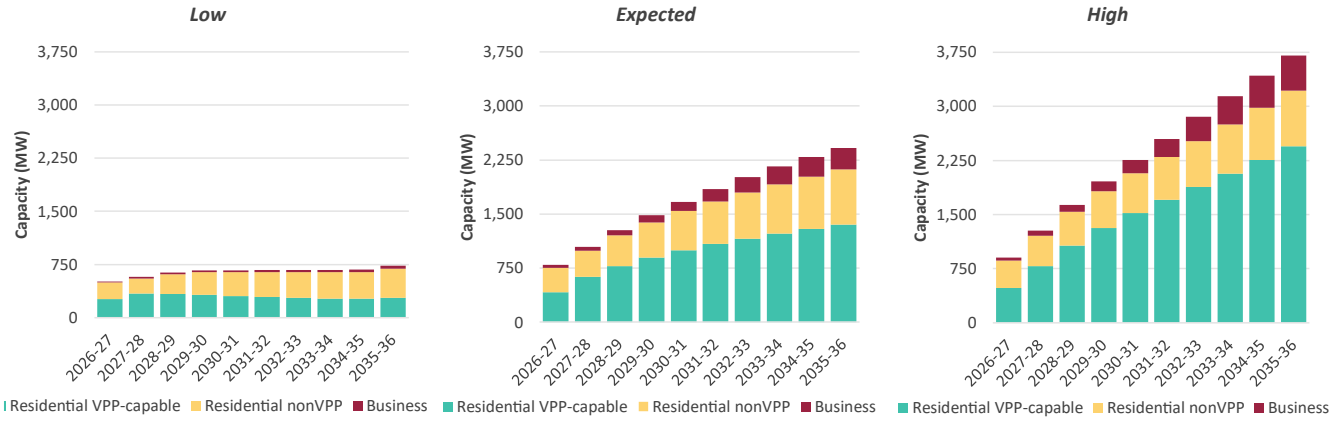


Note: presented on a Capacity Year end basis.

Distributed batteries capacity includes residential and non-residential installations. Residential batteries installed as part of the WA Residential Battery Scheme are required to participate in a VPP arrangement, discussed in Section A1.13). The split of battery capacity into the residential and non-residential segments is presented in **Figure 9**.

[12] See www.wa.gov.au/organisation/energy-policy-wa/wa-residential-battery-scheme and www.dcceew.gov.au/energy/programs/cheaper-home-batteries.

Figure 9 Distributed batteries capacity under three scenarios of the 2026 WEM ESOO spilt by segment, 2026-27 to 2035-36 (MW)



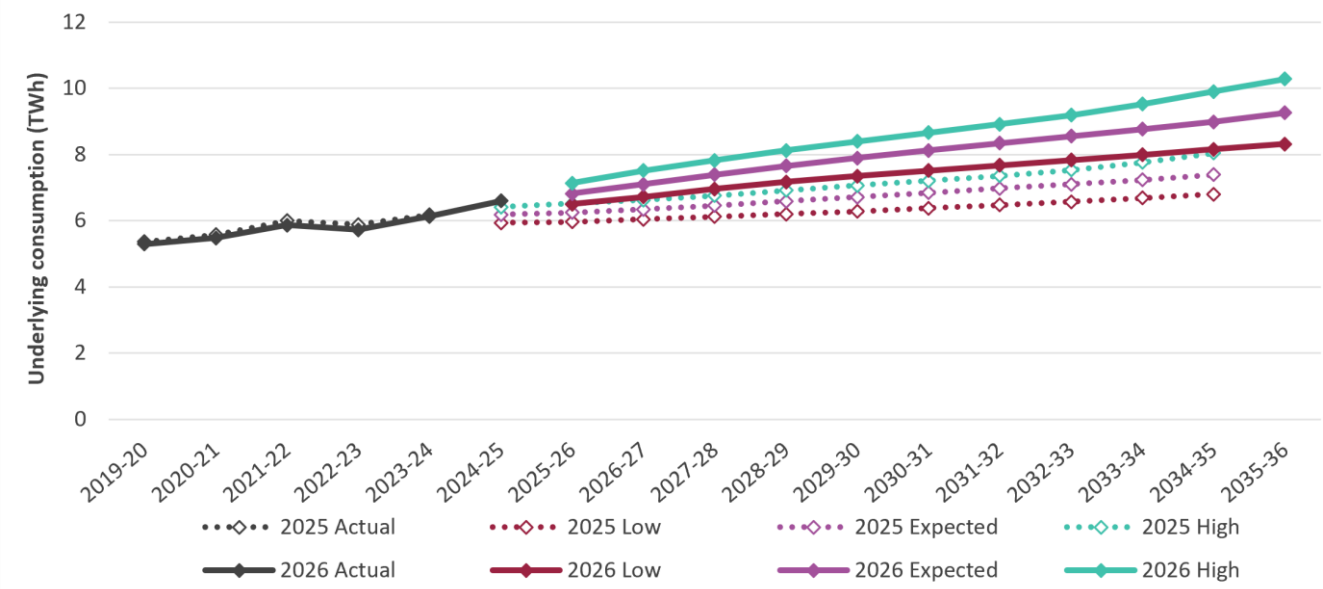
Note: presented on a Capacity Year end basis.

A1.7 Residential consumption

Underlying residential consumption growth is higher across all scenarios in the 2026 WEM ESOO compared to the 2025 forecast. This is driven by a combination of input updates and methodological enhancements, including:

- lower projection of residential connections,
- higher-than-forecast actual consumption data for the most recent year,
- a revised methodology incorporating recently observed customer behaviour trends over the forecast horizon (this is explained further in the following section), resulting in a higher starting point for the 2026 WEM ESOO forecasts,
- revised distributed PV installation estimates that more accurately reflect system replacements, and
- an updated distributed PV forecast that considers greater scenario diversity, particularly for the *Low* scenario, leading to an earlier cross-over in residential delivered consumption scenarios (see **Figure 10**).

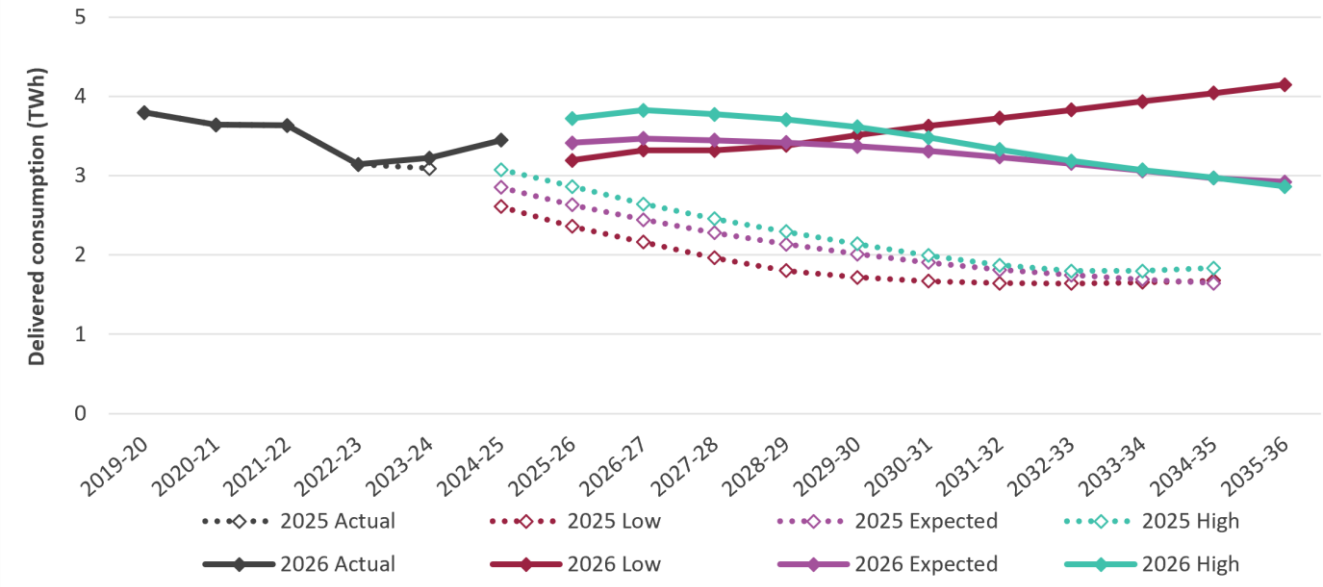
Figure 10 Actual and forecast underlying residential electricity consumption under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36 (TWh)



Note: Values presented in this figure do not include residential battery loss adjustments.

Underlying residential consumption, including the residential EV component of consumption, continues to grow at a higher rate than in the 2025 forecast, driven by updated actuals and the introduction of consumer behaviour trends (see **Figure 12** in Section A1.7.1). However, strong growth in distributed PV more than offsets this increase particularly in the *Expected* and *High* scenarios, resulting in a flat to declining trend in delivered consumption (see **Figure 11** below). By the end of the outlook period, growth in distributed PV generation accounts for 68% of underlying consumption in the *Expected* scenario.

Figure 11 Actual and forecast delivered residential electricity consumption under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36 (TWh)



Note: Values presented in this figure do not include residential battery loss adjustments.

A1.7.1 Residential methodology updated to incorporate recently observed consumer behaviour trends

The residential consumption model includes fitting models that capture customers' underlying electricity consumption response due to temperature using defined heating and cooling day thresholds. This generates an estimate of households' base, cooling and heating load. The total underlying residential consumption forecast is then derived by incorporating a range of additive components before scaling by projected connection growth. Some of these additive components are:

- energy efficiency impact projections,
- influence of EVs,
- residential electrification impacts,
- distributed PV rebound effects, and
- appliance uptake.

Analysis of recent historical data indicates that average household electricity consumption is evolving on a year-on-year basis. As shown in the top-left panel of **Figure 12**, the fitted U-shaped model exhibits a gradual upward shift and a slight narrowing over time. This suggests that both the base load (represented by the horizontal segment of the model) and the temperature-driven responses (namely heating and cooling on the left and right side of the U-shape model, respectively) are increasing annually.

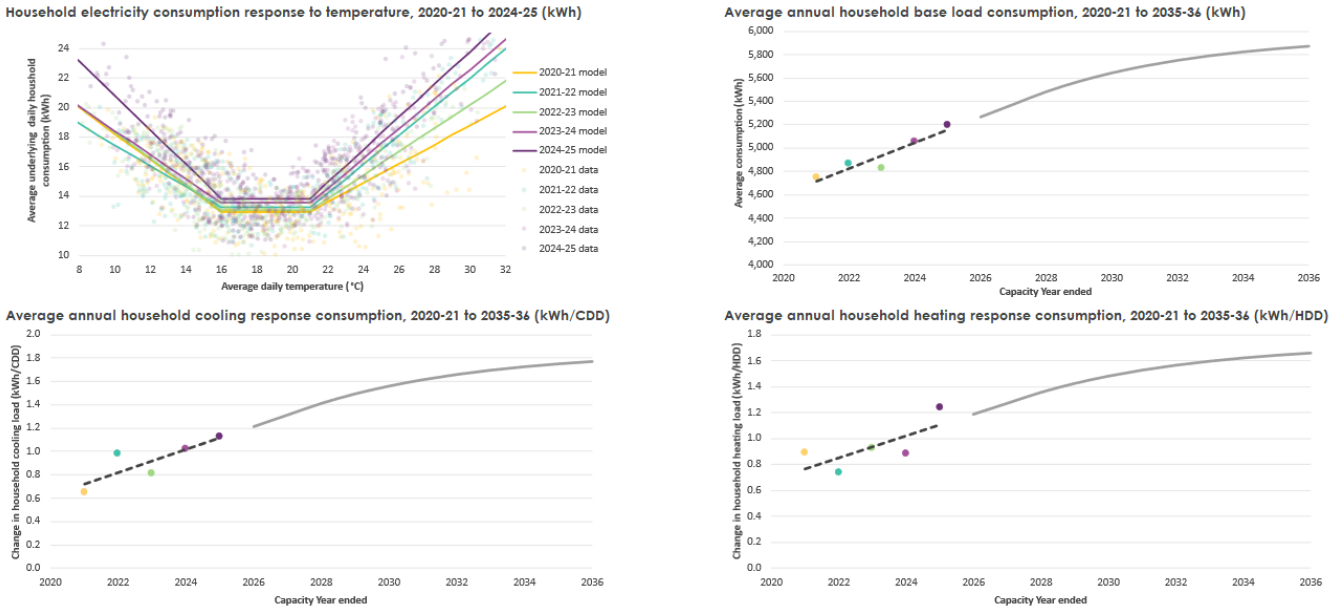
Removing the impact of the above additive components from the historical data confirms the presence of residual consumer behaviour trends, indicating that these effects are not captured by the existing forecast components. Accordingly, the 2026 WEM ESOO introduces an additional component to account for these previously unexplained dynamics.

These behavioural trends are observed across base, cooling, and heating loads, as illustrated in the time series plots in **Figure 12**. For example, the bottom left panel shows the average household's cooling response increased from ~ 0.6 kWh/Cooling Degree Days (CDD)[13] in 2020-21 to ~1.1 kWh/CDD in 2024-2025.

These trends are extrapolated linearly in the near term and are gradually dampened over the long term as other structural forecast components, such as electrification, become more pronounced. Incorporating these trends has led to an uplift in the overall residential forecast and improved alignment with historical underlying consumption trends, as **Figure 12** shows.

[13] Cooling Degree Days refers to the number of degrees that a day's average temperature is above a critical temperature. It is used to account for deviation in weather from normal weather standards. See https://www.aemo.com.au/-/media/files/stakeholder_consultation/consultations/nem-consultations/2024/2024-electricity-demand-forecasting-methodology-consultation/final-determination/electricity-demand-forecasting-methodology_.pdf?rev=683642cd0a264f11b0e5d1c7ef9aefde&sc_lang=en.

Figure 12 Household electricity consumption response to temperature (top left panel) and trends in consumer behaviour across base, cooling and heating load (time series charts)



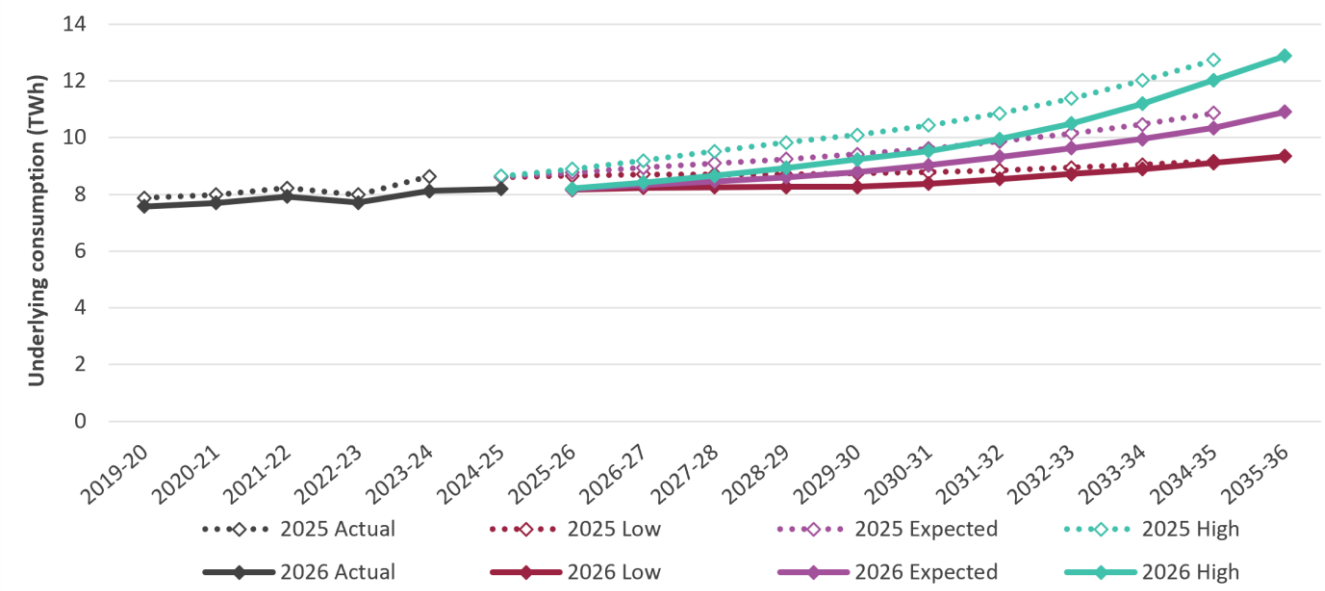
Note: CDD refers to Cooling Degree Days and HDD refers to Heating Degree Days.

A1.8 Business mass market

The business mass market forecast, including the BMM EV component of consumption, in the 2026 WEM ESOO is slightly lower than that presented in the 2025 WEM ESOO. The new forecast incorporates enhancements to the BMM model, resulting in a less pronounced decline in the economy's energy intensity. While this has contributed to an increase in forecast BMM consumption, its impact is outweighed by a downward adjustment to BMM actuals, reflecting the reclassification of certain customers as Industrial or Data Centre loads. Overall, these updates lead to a modest reduction in the underlying BMM forecast compared to the 2025 WEM ESOO as shown in **Figure 13**.

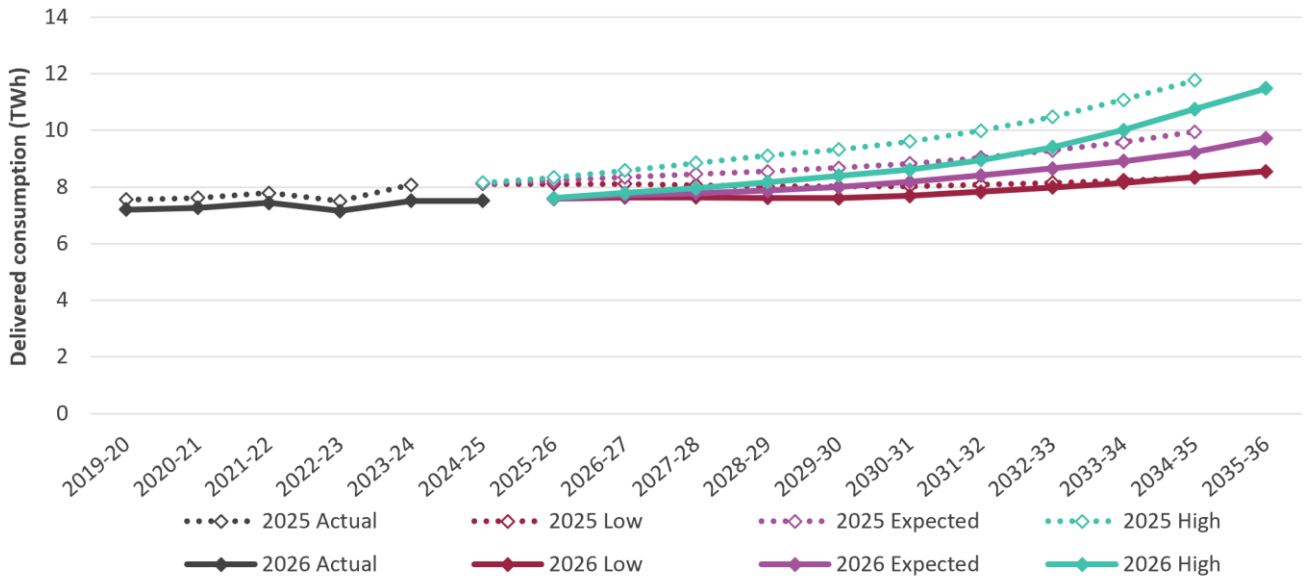
Furthermore, the 2026 WEM ESOO forecast incorporates an upward revision to both historical and projected BMM distributed PV generation, resulting in a further divergence from the 2025 WEM ESOO forecast, as illustrated by **Figure 14**.

Figure 13 Actual and forecast business mass market underlying electricity consumption under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36 (TWh)



Note: Values presented in this figure do not include business mass market battery loss adjustments.

Figure 14 Actual and forecast business mass market delivered electricity consumption under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36 (TWh)



Notes:

- Business mass market delivered consumption includes business EV consumption.
- Values presented in this figure do not include business mass market battery loss adjustments.

A1.9 Large industrial loads

large industrial load are users that consume or are forecast to consume at least 10 megawatts (MW) for a minimum of 10% of the time each year (around 875 hours a year), or at least 50 gigawatt hours (GWh) per year. Large industrial loads consumed approximately 30% of all operational consumption in the SWIS in 2024-25.

For existing large industrial load forecasts, AEMO surveys gain an understanding of their anticipated future consumption and demand forecasts. This survey includes any planned brownfield expansions or closures of existing large industrial loads. AEMO's surveys also seek to identify any future electrification and/or energy efficiency upgrades that are planned for the Facility. The surveys were supplemented by obtaining additional information through interviews. For new greenfield large industrial load forecasts, AEMO undertakes market research to identify prospective projects under development and assess the project development statuses. AEMO has also liaised closely with Western Power, which has provided a detailed forward-looking load connection list, which range from conceptual through to constructed projects.

AEMO's methodology accounts for both long-term government decarbonisation policy targets and the fact that operators usually apply for Environmental Protection Authority (EPA) approvals three to four years in advance of expected Final Investment Decision (FID) for the project in question. For the 2026 WEM ESOO, AEMO refined the new large industrial load project status categories and scoring. For all new large industrial load projects, alternative scoring for pre-FID and post-FID were applied. Pre-FID, project status was scored on a graded scale for each of the status categories described in **Table 1**.

Table 1 Project status scoring for large industrial load pre-Final Investment Decision

Prior	Weighting	Criteria	Score (%)
Western Power Connection	0.143	Connected	100
		Planning	75
		Scoping	50
		Application	25
		No application submitted	0
Environmental Approval	0.143	Stage 1 - Stage 5	20 – 100
		No application submitted	0
Project Phase	0.143	Development	100
		Detailed Design	75
		Front End Engineering and Design	50
		Definitive Feasibility Study	25
		Preliminary Feasibility Study	10
		Screening	0
Planning approval	0.143	Approval received	100
		Application progressing	50
		Application submitted	25
		No application submitted	0
Project Economics	0.143	Strong	100
		Moderate	75
		Marginal	25

Prior	Weighting	Criteria	Score (%)
		Not evaluated	0
Commercial Agreements	0.143	Binding (Offtake) Agreements	100
		Non-binding Heads of Terms	50
		Letter of Interest or Memorandum of Understanding	25
		None	0
Finance	0.143	Financial close	100
		Progressing term sheet	50
		No application	0

Post-Final Investment Decision projects

Following FID, Finance and FID scores are each equal to 100%. Project construction progress remains an outstanding uncertainty and is scored according to the criteria presented in Table 1. Overall, post FID project status score weightings for the three post FID categories i.e. FID, Finance and construction and commissioning, are presented in **Table 2**.

Table 2 Project status scoring for large industrial load post FID

Status Category	Weighting	Criteria	Score (%)
Construction and Commissioning	0.10	Constructed	100
		Commissioning	50
		Progressing	25
		Commenced	0
Finance	0.45	Post-FID/Financial Close	100
		Pre-FID	0
Final Investment Decision	0.45	Positive FID taken	100
		Pre-FID	0

Scoring modifiers

Project status scores are uplifted to reflect higher completion certainty if they receive priority project status from Western Power or State and Commonwealth Governments. These uplifts are:

- Western Power critical project framework[14]: 150% of project status score, and
- Critical Mineral[15] Project: 150% of project status score.

The thresholds for classifying assessed projects are:

- committed capacity: project status score >= 92.5%,
- anticipated capacity: project status score >= 40%, and
- proposed capacity: project status score < 40%.

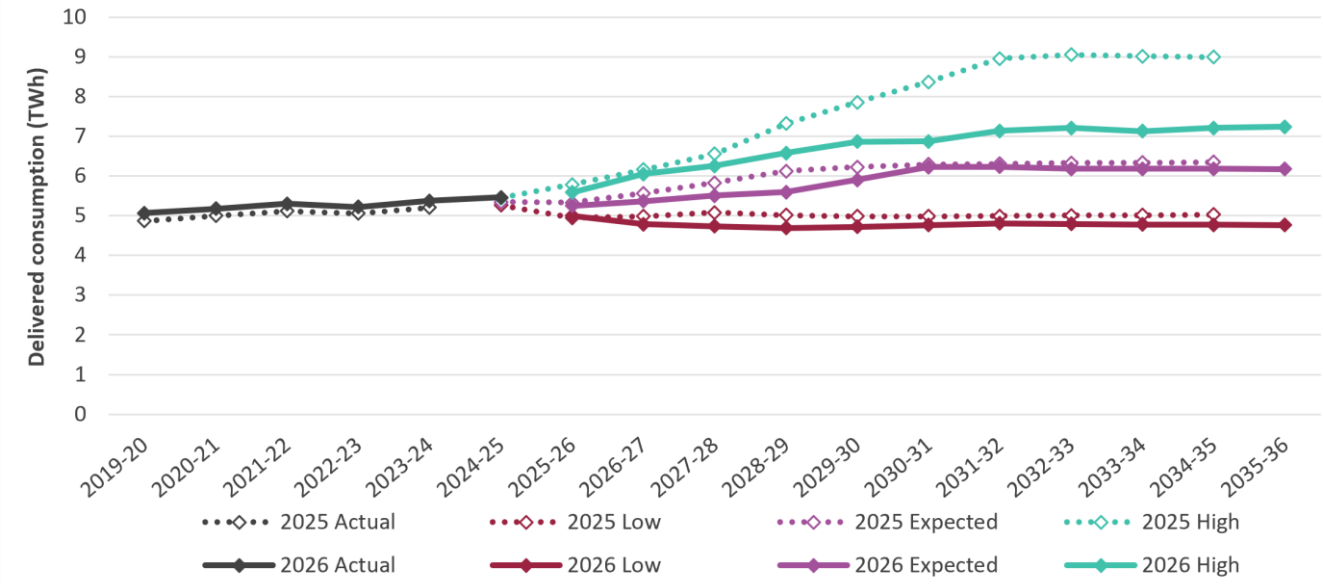
[14] See https://www.westernpower.com.au/resources-education/industry-resources/large-industry-customers-and-generators/critical-project-framework/.

[15] See https://www.industry.gov.au/mining-oil-and-gas/minerals/critical-minerals.

Large industrial load forecasts

Figure 15 compares existing and new large industrial load consumption forecasts for the *Low*, *Expected*, and *High* scenarios from the 2025 and 2026 WEM ESOOs, and shows actual large industrial load consumption from 2019-20 to 2024-25. Large industrial load consumption shown here does not include electricity consumption from hydrogen projections and electrification of large industrial load[17].

Figure 15 Actual and forecast large industrial load delivered electricity consumption under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36 (TWh)



Despite many changes from year to year, particularly from 2025 to 2026, the aggregate large industrial load outlook is subject to only a small downward revision[16]. Key changes include:

- Several new sites have been added to the large industrial load category, resulting in higher actuals.
- Some operators have advised of immediate mothballing or closure, and some have advised a projected closure date later in the horizon.
- Some previously considered industrial developments are no longer proceeding or have advised later expected start dates.
- Several have advised of an expectation to build behind the meter distributed PV and wind generation, with an associated reduction in requirement for delivered WEM consumption.
- Some industrial operators have advised that they intend to stop or reduce behind-the-meter generation, which results in an increased consumption outlook.
- Additionally, to address a persistent over-forecasting that has been observed between actuals and survey responses, an outage factor has been incorporated that reduces the large industrial load consumption forecast.

[16] For numerous large industrial loads with complex generation arrangements behind the meter in the SWIS, AEMO is unable to forecast underlying consumption, hence underlying is set to equal delivered for these customers. True underlying consumption is therefore higher in the SWIS than represented.

A1.9.1 Large industrial load consumption forecast calibration utilising outage factors

Of the many variables driving the large industrial load forecasts, the outage factor has been found to be one of the more significant. AEMO conducted an analysis of accuracy of information received from industrial operators covering respective energy consumption. This analysis has revealed that the information received from operators over the last four years has consistently exceeded actual consumption, in some cases significantly. Some factors contributing to actual consumption being below survey estimates include planned and unplanned load outages, poorer economic conditions or commodity prices than anticipated, and expansions and developments occurring behind schedule or being cancelled.

To address this issue, AEMO has introduced an outage factor that is applied to the large industrial load forecasts. This factor has been developed by taking the survey-based consumption from the last four years surveys, and aggregating the past projections and actuals to estimate outage impacts, ensuring results are not unduly affected by new developments or anomalous data points.

For the existing operators, a mean outage factor of 5% has been applied to the *Expected* scenario, and to reflect year-on-year variability, the *Low* scenario would apply a 7.5% outage factor and the *High* scenario a 2.5% outage factor. As a result of these, the 2026 forecast is a slightly lower than the 2025 forecast.

A1.10 Energy efficiency savings forecast to grow strongly throughout the outlook period

The forecasts for energy efficiency savings[17] reflect the potential role of energy efficient choices by considering effects on energy savings attributable to investment in new technologies, buildings, and processes, and uptake of energy-efficient appliance and equipment in business and residential sectors[18]. The forecasts incorporate market-led efficiency improvements and federal and state policy measures, with progressively stronger future adjustments applied from the *Low* to *Expected* and *High* scenarios, consistent with the 2025 IASR scenario narratives. Energy efficiency savings have been fully incorporated into the residential and BMM forecasts.

The energy efficiency forecasts considered these government measures aimed at lowering energy consumption:

- building energy performance requirements contained in the National Construction Code (NCC) 2022,
- the Greenhouse and Energy Minimum Standards (GEMS) program of mandatory product labelling requirements and/or minimum energy performance standards for different classes of appliances and equipment, and
- building rating and disclosure schemes for existing commercial offices, including the Commercial Building Disclosure (CBD) and the National Australian Built Environment Rating System (NABERS) for offices.

Figure 16 presents the total additional energy efficiency savings forecast for the residential and BMM sectors relative to the base year; therefore, values begin from a zero baseline at the start of the forecast period. Energy efficiency savings for large industrial loads are included within their survey responses and are therefore captured in the large industrial load forecast results (in Section A1.9). Overall, the pace of savings from energy efficiency improvements is similar to the 2025 WEM ESOO following rebasing the values.

[17] Energy efficiency relates to how much energy is consumed to perform a task whereby policy seeks to improve this efficiency. For detail, see https://aemo.com.au/-/media/files/major-publications/isp/2025/stage-2/2025-energy-efficiency-forecasts-final-report.pdf?la=en.

[18] Fuel-switching, including electrification, will generally deliver changed energy efficiency, but it is excluded to avoid double-counting (see **Section A1.4** for details about electrification).

Figure 16 Forecast total energy efficiency savings in the business mass market and residential sectors, under three scenarios from the 2025 and 2026 WEM ESOOs, 2024-25 to 2035-36 (TWh)



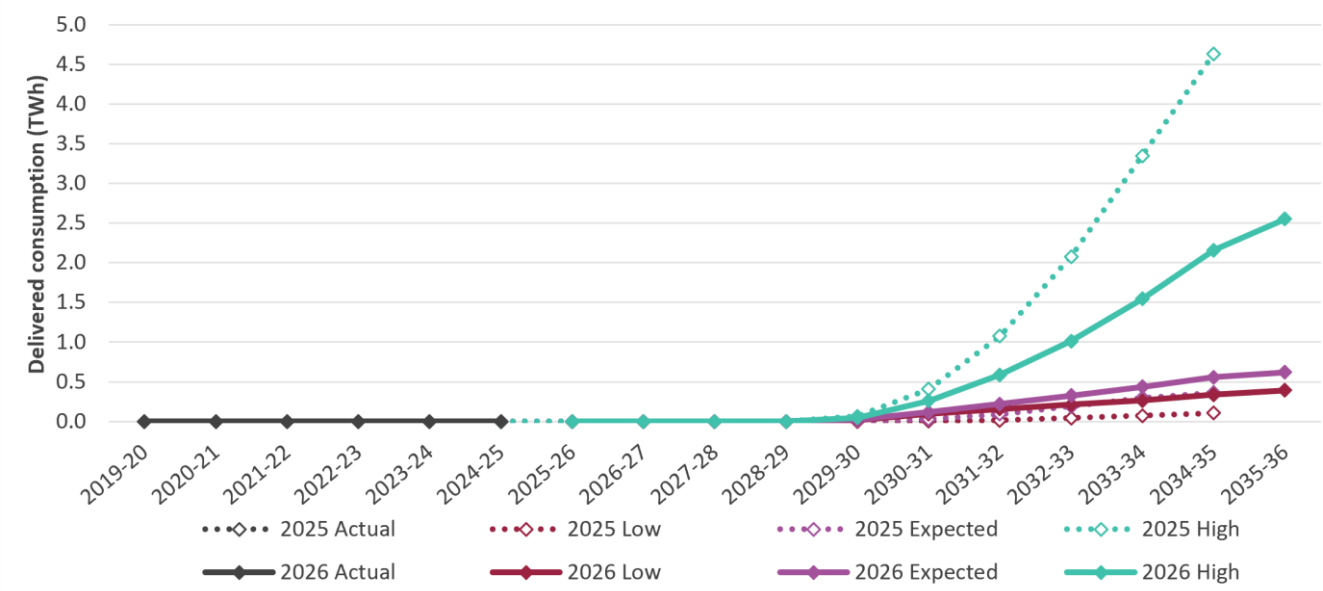
A1.11 New industries: Forecast growth in hydrogen production is significantly reduced compared to the 2025 WEM ESOO

AEMO's hydrogen forecast captures electricity consumption from electrolysers used to produce hydrogen from renewable energy sources for domestic use and for applications such as green steel production.

The 2026 WEM ESOO hydrogen-related electricity consumption forecasts in the *Expected* and *Low* scenarios are comparable to the 2025 WEM ESOO forecasts, whereas the 2026 WEM ESOO *High* scenario is considerably lower.

This is largely driven by changes in scenario assumptions since the 2025 WEM ESOO, which applied the draft 2025 IASR *Green Energy Exports* scenario with faster hydrogen uptake. By comparison, the 2026 WEM ESOO applies the final 2025 IASR *Accelerated Transition* scenario (previously *Green Energy Industries*), which reflects slower hydrogen uptake and excludes hydrogen exports across the forecast period.

Figure 17 Actual and forecast new industries delivered electricity consumption under three scenarios, from 2025 and 2026 WEM ESOOs, 2019-20 to 2035-36 (TWh)



A1.11.1 Approach for hydrogen project demand modelling during maximum peak demand

The economics of hydrogen production is highly influenced by the utilisation factor of the electrolysers, defined as the proportion of hydrogen produced in a year, relative to the maximum potential annual production from the electrolysers.

The contribution to demand from hydrogen production was modelled to be curtailed by 90%[19] during peak and high demand periods to reflect cost efficient production. Peak and high demand periods are defined as the threshold occurring when SWIS operation demand exceeds the 70% of the SWIS load duration curve. This reflects a hydrogen production utilisation factor of 70%[20] (see **Figure 32**, Section 4.2.2 of the 2026 WEM ESOO report).

Curtailed hydrogen production occurs during summer and winter peak demand periods (see Sections 2.3.1 and 2.6 of the 2026 WEM ESOO report) as these exceed the threshold operational demand level. When SWIS operational demand falls below the threshold, which includes periods of minimum demand, full hydrogen loads apply.

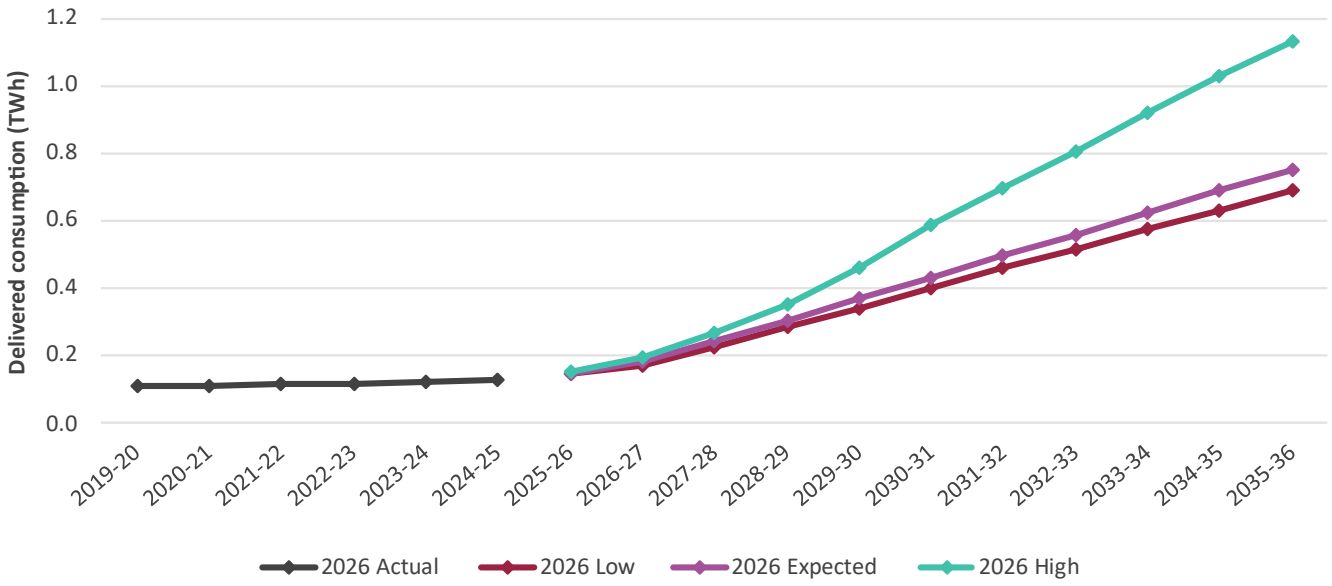
A1.12 Data centres

While data centre loads were previously included under the BMM category, the 2026 WEM ESOO introduces them as a distinct, standalone category. The main locations for data centre development in Australia are in Sydney and Melbourne, with less intense interest in locations outside of these two cities. Therefore, the forecast for data centre consumption in the SWIS is moderately low.

[19] 90% based on AEMO assessment of hydrogen industry operations.

[20] See *2025 Inputs, Assumptions and Scenario Report*, page 73, at https://www.aemo.com.au/-/media/files/stakeholder_consultation/consultations/nem-consultations/2024/2025-iasr-scenarios/final-docs/2025-inputs-assumptions-and-scenarios-report.pdf?rev=63268acd3f044adb9f5f3a32b6880c27&sc_lang=en.

Figure 18 Actual and forecast data centres delivered electricity consumption under three scenarios, from 2026 WEM ESOO, 2019-20 to 2035-36 (TWh)



Consumption actuals have been developed by identifying all existing data centres in the SWIS, and projections have been developed using the AEMO EDFM methodology[21] applied in the National Electricity Market (NEM) 2025 ESOO.

The data centre consumption forecast is derived as the average of the project view and the economic view. The project view significantly uplifts the average.

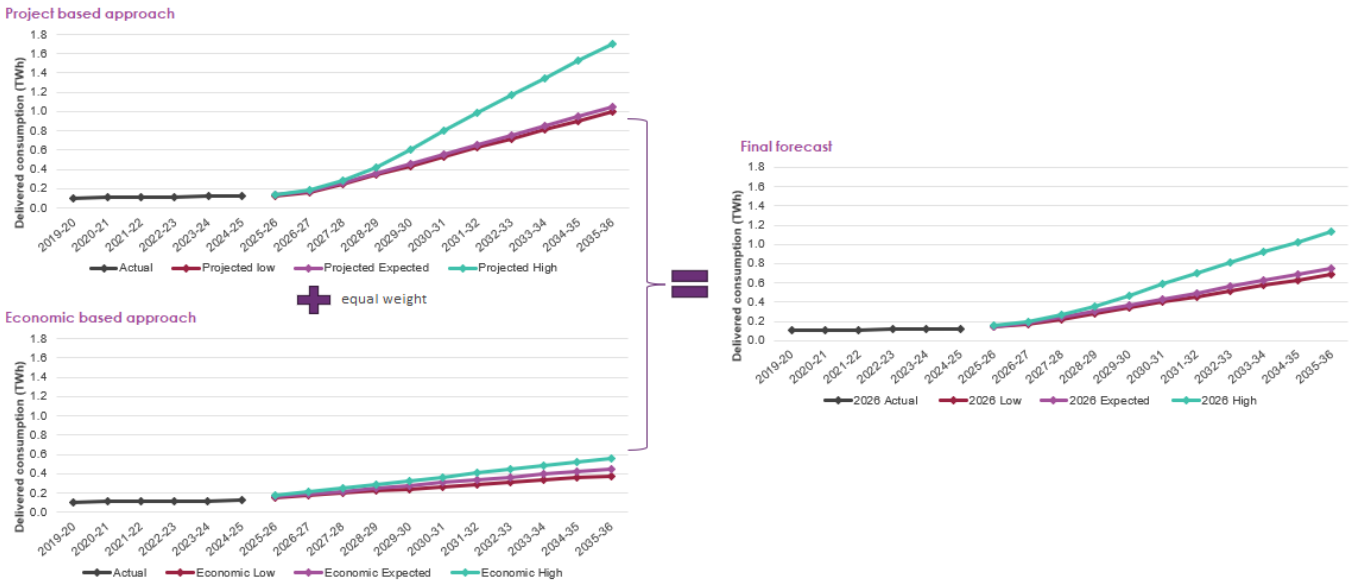
This methodology[21] takes two sets of inputs:

- **Project based forecast** – the forecast is based on new specific committed or anticipated projects together with existing operators. Information was gathered directly through surveys from larger existing operators and information about new projects seeking to connect to the grid was sourced from Western Power. Only projects that are significantly advanced in the grid connection process are considered in the project view. While there are a large number of projects in the very early stages of seeking to connect to the SWIS, only a small number of projects are in advanced stages.
- **Economic based forecast** – underlying economic demand for data centre services in Australia was procured from Oxford Economics Australia[22] and is based on a variety of different projections of underlying economic demand for data centre services. In the 2026 WEM ESOO, the economic view is forecasting a stronger growth rate than has occurred in the past, which reflects potential from early-stage projects coming into service as time progresses.

[21] See https://www.aemo.com.au/-/media/files/stakeholder_consultation/consultations/nem-consultations/2024/2024-electricity-demand-forecasting-methodology-consultation/final-determination/electricity-demand-forecasting-methodology_.pdf?rev=683642cd0a264f11b0e5d1c7ef9aefde&sc_lang=en.

[22] See https://www.aemo.com.au/-/media/files/stakeholder_consultation/consultations/nem-consultations/2024/2025-iasr-scenarios/final-docs/oxford-economics-australia-data-centre-energy-consumption-report.pdf?la=en.

Figure 19 Data centre forecasts methodology: Blending approaches – project based and economic based forecasts under three scenarios, 2026 WEM ESOO, 2019-20 to 2035-36 (TWh)



A1.13 The impact of distributed batteries on unscheduled operational demand

Distributed batteries, together with distributed photovoltaics (PV) and electric vehicles (EV), are classified as distributed energy resources (DER). DER are smaller-scale, consumer-owned electrical devices that generate or store electricity, located "behind-the-meter" and used within residential or business premises to self-supply co-located loads.

DER impact the broader, "front-of-the-meter" power system, through their passive (uncoordinated) or coordinated behaviour. **Passive (uncoordinated) behaviour** means operation to meet the energy needs of an individual behind-the-meter energy system, without regard for and response to the operating conditions of the broader, front-of-the-meter power system, and the coinciding electricity market outcomes. In contrast, **coordinated behaviour** means operation with consideration for, and in response to, the broader power system and electricity market conditions. The coordination involves hardware and software solutions to enable remote communications and control of the distributed devices, and turns them into an aggregated and controllable resource, that is, a virtual power plant (VPP).A VPP can support the reliability and security of the broader power system by:

- reducing demand in peak periods, which helps reduce the risk of unserved energy,
- increasing demand in troughs, which helps maintain the Minimum Demand Threshold for inertia and reactive power management purposes, and
- providing Essential System Services (ESS) for frequency and voltage control purposes.

Coordinated DER in the SWIS builds on the Western Australia Government's DER Roadmap[23], related smaller-scale trials such as Project Symphony and Project Encore, and Project Jupiter (launched in early 2025 and currently ongoing) to support large-scale integration of DER through VPPs.

[23] See https://www.wa.gov.au/government/distributed-energy-resources-roadmap.

The 2026 WEM ESOO *Expected* scenario forecasts a strong uptake of residential batteries in the SWIS, with a significant share of VPP-capable batteries. This uptake is driven by the WA Residential Battery Scheme, which offers a rebate or a no-interest loan to purchase a home battery, and requires that the batteries participate in the VPP for at least two years.

Considering the above, the 2026 WEM ESOO models the impact of distributed residential batteries on the SWIS unscheduled operational demand as described in Section A1.13.1 below.

A1.13.1 Modelling the impact of passive batteries and batteries coordinated through a virtual power plant

The impact of passive batteries on SWIS unscheduled operational demand is modelled in line with AEMO's *Electricity Demand Forecasting Methodology*[24] and captures at least two broad types of battery operation:

- solar shift, where the battery will charge when excess solar PV generation is available and discharge whenever solar PV generation is insufficient to cover household demand, and
- time of use, where the battery is optimised to take advantage of a time of use tariff, topping up charge at off peak times to maximise avoidance of peak time tariffs.

The 2026 WEM ESOO models the impact of coordinated distributed batteries (VPP) on the SWIS unscheduled operational demand for the first time[25]. The modelling was conducted as a separate post-process after the modelling of passive batteries.

The modelling of coordinated distributed batteries focuses on residential batteries and considers the Synergy-operated VPP. It simulated the VPP's discharge during peak demand periods to reduce exposure to the cost of the Individual Reserve Capacity Requirement (IRCR). The behaviour of the VPP under the AEMO Non-Co-optimised Essential System Services (NCESS), and Western Power Network Support Services (NSS) was not modelled, as discussed further below in **Table 3**. VPP capacity associated with AEMO NCESS Peak Demand Service was included in the supply-side supply side modelling.

In the SWIS, customers that consume no more than 50 MWh of electricity each year cannot be supplied by a retailer other than Synergy. These customers are referred to as 'non-contestable customers'.

Synergy is the 'Parent Aggregator' for non-contestable customers, and any third party wishing to provide services to the network operator or market through non-contestable DER will be required to form agreements with Synergy.

In this document, '"the VPP" therefore refers to the VPP coordinated by Synergy as the Parent Aggregator.

The modelling of the VPP was informed by discussions with relevant stakeholders on the intended operation of the VPP and AEMO's research on VPPs in other jurisdictions[26], and built on the in-house draft methodology presented by AEMO in the FRG on 30 March 2026[27].

[24] See www.aemo.com.au/-/media/files/stakeholder_consultation/consultations/nem-consultations/2024/2024-electricity-demand-forecasting-methodology-consultation/final-determination/electricity-demand-forecasting-methodology_.pdf.

[25] In the 2025 WEM ESOO, batteries coordinated as a VPP were modelled on the supply side, with the same methodology as applied to large-scale storage. See www.aemo.com.au/-/media/files/electricity/wem/planning_and_forecasting/esoo/2025/ey---2025-wem-esoo-reliability-assessment-methodology-report.pdf.

[26] Examples of coordinated DER responses at scale include a 29 July 2025 event in California, where aggregated dispatch from more than 100,000 batteries reduced peak demand by an average of 539 MW between 19:00 and 21:00, and a 20 June 2024 event in New England, where aggregated dispatch of batteries, EVs and residential loads reduced peak demand by an average of 375 MW between 17:00 and 20:00. See www.brattle.com/wp-content/uploads/2025/12/The-Demand-Side-Grid-Support-Program-An-Assessment-of-Scale-and-Value-December-2025-Update.pdf and www.wbur.org/news/2024/08/28/virtual-power-plants-eversource-massachusetts-batteries-ev-chargers.

[27] See www.aemo.com.au/-/media/files/stakeholder_consultation/working_groups/wa_meetings/wem-esoo-frg/2026/wem-esoo-frg---meeting-papers---30-march-2026.pdf.

High-level methodology steps

The modelling of the VPP's impact on peak demand was performed in accordance with the following steps, explained in more detail in **Table 3** and illustrated in **0**:

1. **Develop pre-VPP demand profiles** – develop forecasts of time-sequential, per-interval profiles of unscheduled operational demand[28] in line with AEMO's *Electricity Demand Forecasting Methodology*, where all distributed batteries (residential and business) operate in a passive way.
2. **Develop adjusted demand profiles** – for peak demand periods, adjust the pre-VPP demand profiles by excluding the impact of VPP-capable batteries operated passively; this adjustment increases demand.
3. **Model the coordinated discharge of the VPP** – using the adjusted demand profiles, model the coordinated discharge of the VPP-capable batteries to target periods of highest demand (VPP activation event), and the discharge of the VPP-capable batteries returning to passive operation following a VPP activation event.
4. **Derive post-VPP demand profiles** – develop post-VPP demand profiles by subtracting the modelled discharge of the VPP from the adjusted demand profiles; this is done while maintaining the same-interval logic[29].

The post-VPP demand profiles were used by AEMO to calculate the VPP's impact on annual peak unscheduled operational demand and perform the Limb A assessment. For each modelled year, the impact was calculated as the difference between the annual peak demand modelled in the post-VPP demand profiles, and the annual peak demand modelled in the pre-VPP demand profiles, noting that these may occur:

- on the same day, in the same interval (for example, 18:00 on 17 December),
- on the same day, but in different intervals (for example, 16:30 and 18:00 on 17 December),
- on different days, in the same interval (for example, 18:00 on 17 December, and 18:00 on 6 January), and
- on different days and different intervals (for example, 18:00 on 17 December, and 16:30 on 6 January).

The post-VPP demand profiles were also used by AEMO in the Limb B and sub-regional assessments, the Limb C assessment, and other components of the SWIS reliability assessment framework, such as the determination of the Electric Storage Resource (ESR) Duration Requirement, the ESR Duration Requirement Uplift, and the Availability Curves.

Virtual power plant modelling inputs, assumptions and implementation

The modelling of the VPP's impact on peak demand used the inputs and assumptions summarised in **Table 3**. Draft inputs and assumptions were presented to the FRG on 30 March 2026[30], and further refined with Synergy to reflect Synergy's current view of the intended operation of the VPP.

[28] These are time-sequential series of 17,520 data points per year (17,568 in leap years). They represent unscheduled operational demand, that is operational demand exclusive of the impact of operational withdrawals (charging of large-scale storage, or demand reduction from large loads), or the impact of the Emergency Solar Management scheme (curtailment of distributed PV).

[29] This means subtracting the coordinated discharge of the VPP modelled from the adjusted demand modelled for the same interval, for example, 18:00 on 17 December 2028.

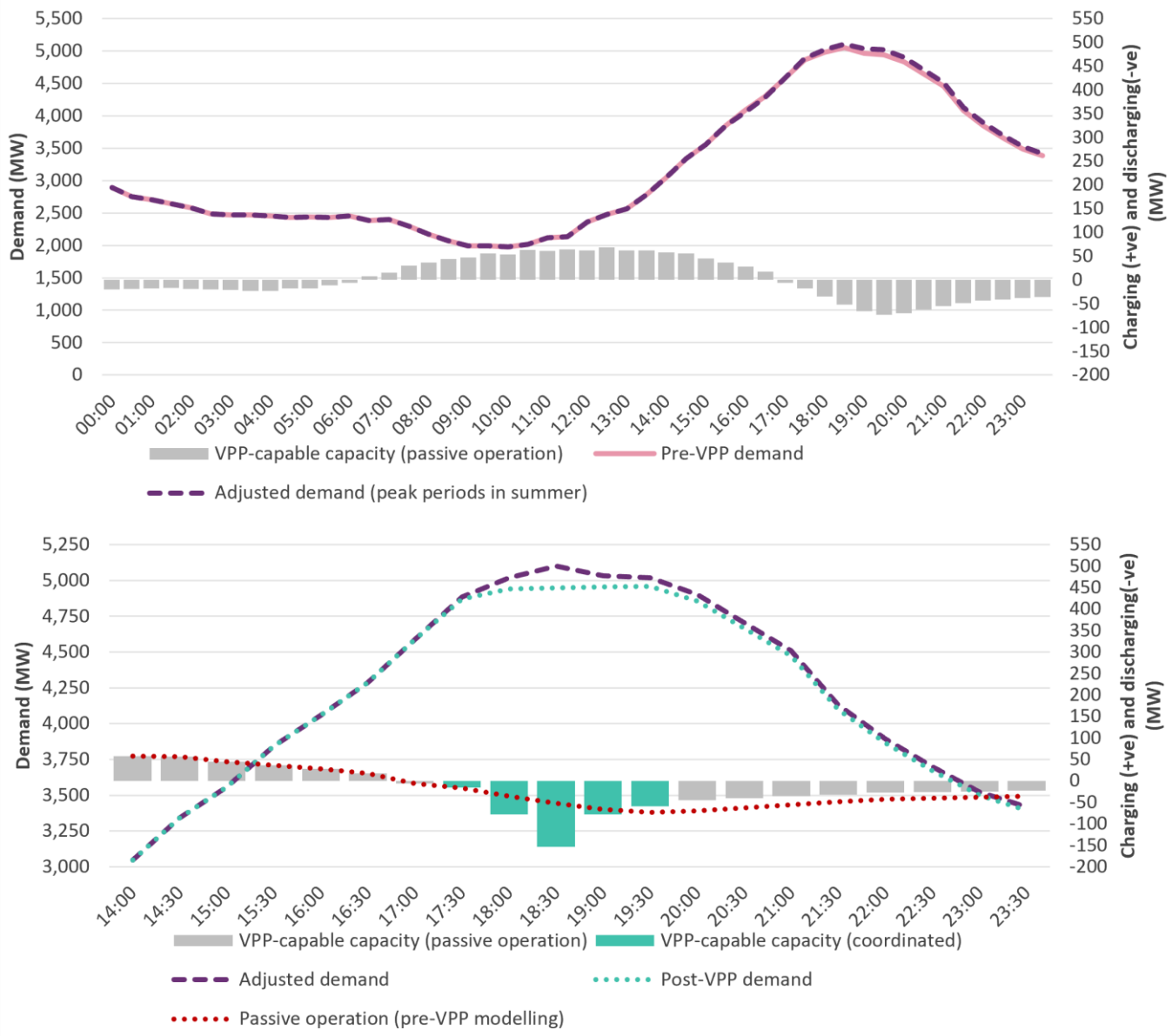
[30] See www.aemo.com.au/-/media/files/stakeholder_consultation/working_groups/wa_meetings/wem-esoo-frg/2026/wem-esoo-frg---meeting-papers---30-march-2026.pdf.

Table 3 Inputs and assumptions used in the modelling of the impact of the virtual power plant

Item	Details
Forecast uptake trajectories of the power capacity (MW) and energy capacity (MWh) of residential distributed batteries	The modelling uses uptake trajectories of non-VPP and VPP-capable residential distributed batteries for the *Expected* scenario, as presented in **Appendix A1.13**. The data includes end-of-month power capacity (MW) and energy capacity (MWh) of batteries. The energy capacity of the batteries includes the impact of degradation.
Timing and duration of the VPP activation events	A VPP activation event is a set of consecutive Trading Intervals over which the VPP is expected to discharge. For each modelled year, the VPP's discharge was simulated for daily peak demand periods across the 30 highest-demand days in summer (December to March). The timing of activation events was assumed as 17:30-19:00 (2 hours) in 2026-27, growing to 2.5 hours by 2028-29 (17:00-19:00 and 17:30-19:30), and 4 hours (16:30-20:00) by 2035-36.
Perfect foresight and predictability of IRCR intervals	The VPP modelling process uses unscheduled operational demand profiles as inputs, which assumes perfect foresight of peak demand periods and informing activation of the model to meet IRCR intervals. In real-life operations, the predictability of IRCR intervals is likely to have a margin of error. In the 2027 WEM ESOO, AEMO will consider improvements to the modelling process to include the uncertainty around IRCR intervals predictability.
The use of the VPP for AEMO NCESS Peak Demand Service, Western Power NSS and IRCR cost reduction	The modelling focussed on the behaviour of the VPP to achieve IRCR cost reduction. The potential use of the VPP for Western Power NSS[A] was considered but not modelled in this WEM ESOO. VPP capacity associated with AEMO NCESS Peak Demand Service is included in the supply-side modelling. The AEMO NCESS Peak Demand Service contract ends in October 2027. As such, it does not impact 2028-29. To derive the VPP-capable capacity for IRCR reduction in the summer period, AEMO subtracted NCESS and NSS related capacity from the forecast capacity of VPP-capable batteries. In years following 2028-29, AEMO assumed a linear growth of the NSS capacity. The forecast capacity of VPP-capable batteries across the 2028-29 summer exceeds 680 MW (monthly average across December to March), of which approximately 640 MW is forecast to be used for IRCR reduction[A]. This is significantly greater than the capacity that it currently contracted to be used for NSS (approximately 44 MW[B], which is due to expire in March 2029). AEMO notes that Western Power recently published a call for submissions to deliver NSS to ensure feeder loading remains within planning limits for the period 1 December 2027 to 31 March 2030[C]. AEMO will engage with Western Power to understand the role VPPs will play in delivering NSS and will consider including the impact of this in future WEM ESOOs.
Availability factor	To account for uncertainties around remote control and operational behaviour due to the early stage of the VPP's development, AEMO assumed a 50% availability factor. This means that 50% of the VPP-capable capacity is modelled as coordinated (e.g., around 320 MW in summer 2028-29), while the rest is modelled in a passive way.
Round-trip efficiency	The modelling assumes an 85% round-trip efficiency of the batteries, in line with AEMO's *2025 Inputs and Assumptions Workbook*[D].
State of charge (SoC)	The modelling assumes that the VPP's SoC at the beginning of an activation event is at least 90%. The VPP is assumed to accumulate this charge primarily from surplus on-site rooftop PV. As advised by Synergy, the VPP may top-up its charge from the broader SWIS, if needed. The discharge of the VPP across an activation event stops when the state of charge reaches 10%.
Discharge profile	Across an activation event, the modelling simulates the discharge of the VPP to "shave and smooth" the profile of the underlying operational demand, and to prevent interval-to-interval "swings" of demand. This means that the VPP's discharge power varies from interval to interval of an activation event rather than remaining constant. In intervals immediately following the discharge of the VPP, batteries are modelled to return to passive operation and discharge any energy remaining after the activation of the VPP.

A. Calculated as the forecast capacity of 680 MW less the maximum capacity for NSS, that is 44 MW.
B. In January 2026, Western Power published a "Draft NCESS Service Specification". See www.wa.gov.au/system/files/2026-01/ncess_-_trigger_submission_-_swis_network_support_-_public.pdf and www.westernpower.com.au/490b69/siteassets/documents/rois/draft-ncess-service-specification-swis-network-support.pdf.
C. AEMO notes that a "Final NCESS Service Specification" was published by Western Power on 27 May 2026. AEMO's modelling of the VPP's impact was concluded at the beginning of May 2026. The final specification also added an NSS requirement for 2029-30, which was not included in the January 2026 draft specification. See www.westernpower.com.au/49bf97/siteassets/documents/rois/ncess-final-service-specification-swis-network-support-2027-30.pdf.
D. See www.aemo.com.au/-/media/files/stakeholder_consultation/consultations/nem-consultations/2024/2025-iasr-scenarios/final-docs/2025-inputs-and-assumptions-workbook.xlsm.

Figure 20 Illustration of the approach to modelling the impact of the virtual power plant on peak demand (MW)



Virtual power plant modelling limitations

Historically, DER coordination in the SWIS has been explored in smaller-scale trials such as Project Symphony and Project Encore, which involved around 900 DERs across 500 homes and businesses.

The 2026 WEM ESOO *Expected* scenario forecasts a strong uptake of residential batteries in the SWIS, with VPP-capable batteries forecast to exceed 126,000 systems in 2028-29.

To support large-scale integration of DER across the SWIS through VPPs, Project Jupiter was launched in early 2025 and is currently ongoing.

The 2026 WEM ESOO's first-time modelling of the VPP's impact on unscheduled operational demand is therefore performed against the backdrop of limited historical data and operating experience, as well as uncertainties around remote control and future operational behaviour of the VPP.

Limitations of this year's modelling also include data representing the passive behaviour of batteries. Compared to passive charge-discharge profiles used by AEMO's *Electricity Demand Forecasting Methodology*, estimates of SWIS-specific data indicate a different timing and a higher contribution of passive batteries to reducing demand in peak periods. To reflect this, AEMO has uplifted the impact of passive batteries, which acts to increase the total modelled impact of residential batteries (passive and coordinated) on demand. The uplift was assessed based on sample data collected for the SWIS relative to charge-discharge profiles used by AEMO's *Electricity Demand Forecasting Methodology*.

As noted above in **Table 3**, the VPP modelling process uses unscheduled operational demand profiles as inputs, which assumes perfect foresight of peak demand periods and informs activation of the VPP for IRCR intervals. In real-life operations, the predictability of IRCR intervals in unlikely to be 100% accurate.

AEMO expects to develop and refine the VPP modelling methodology by building on real-time operational data for the SWIS and learnings of Project Jupiter, considering improvements to the modelling process to include the uncertainty around IRCR intervals, and any further VPP use cases.

A2. Assessment of supply projects

A2.1 New project status evaluation methodology

AEMO assesses all projects for the development of 2026 WEM ESOO supply modelling based on information it received through:

- responses to the 2026 Long Term Projected Assessment of System Adequacy (PASA) formal information requests (FIR) from Market Participants and Western Power as part of the 2026 WEM ESOO development process, and
- submissions for 2026 Expressions of Interest (EOI)[31] for Certified Reserve Capacity (CRC).

The projects can be at various phases of its development, such as planning phase, in the approvals phase, or in the construction phase at the time of submitting information to AEMO. All projects from these responses are assessed based on a project status evaluation methodology.

New projects are those that are planned to connect in the SWIS and are not yet operational. New projects are classified into **committed**, **anticipated** and **proposed** capacity categories based on AEMO assessment criteria described below, evaluating project maturity and likelihood of delivery of individual projects.

New projects that received Capacity Credits[32] through to 2027-28 or contracted for the 2025-27 Peak Demand NCESS are considered as committed capacity and therefore are exempted from further evaluation to assess their status as new projects. For these contracted capacities, the timing of the projects is monitored through Progress Reports (as submitted by Market Participants to AEMO) and project updates received from Western Power.

Five criteria applied to rank each of the new projects are defined below. Corresponding to the status of a project against each of these criteria, AEMO applies a percentage score, detailed in **Table 4** below.

1. **Land approval** – for a project to proceed, the proponent needs to have completed the purchase, acquisition or settlement of land. AEMO assigned a scoring-based land approval status.
2. **Planning approvals** – for a project to proceed, the proponent needs to receive all required environmental and development approvals from Western Australia's EPA and concerned Western Australian authorities, respectively. AEMO assigned a scoring-based planning approval status.
3. **Western Power connection access** – approval for connection access is obtained by AEMO directly from Western Power. AEMO assigned a scoring-based status following information received from Western Power.
4. **Finance approval by proponent** – Final Investment Decision (FID) is taken by a proponent once all internal studies and planning have been completed, the environmental approvals are in place, and the commercial work (for example, fuel and sales agreements) has been finalised. Projects can be at different stages towards achieving the FID when submitting a FIR or an EOI. AEMO assigned a scoring-based project finance status.

[31] See 2026 Expression of Interest Summary Report, 2026, at https://www.aemo.com.au/-/media/files/electricity/wem/reserve_capacity_mechanism/eoi/2026/2026-expression-of-interest-summary-report.pdf.

[32] Capacity Credits presented here refer to the Peak Reserve Capacity determined in the corresponding Reserve Capacity Cycle unless otherwise specified.

5. **Construction** – for a project to proceed with construction, the proponent needs to have received all necessary approvals including building permit from associated governing authorities to be compliant for building construction. AEMO assigned a scoring-based construction status.

For each project, a total project status score (in percentage) has been calculated using the weightings outlined in **Table 4**. The threshold scores for classifying projects are:

- committed capacity if a project status score >= 80%,
- anticipated capacity if a project status score >= 60%, and
- proposed capacity if a project status score < 60%.

This methodology has been expanded from previous WEM ESOOs to consider broader development criteria in assessing and categorising new projects. The 2026 WEM ESOO new project scoring methodology provides an structured assessment of project development maturity. AEMO applies judgement where a single criterion, such as network access or finance approval, materially affects the likelihood or timing of project delivery which aligns closely with the NEM approach. AEMO has also varied the names of the capacity classification from 'committed', 'probable' or 'proposed'. These changes better align with the NEM classifications of 'committed', 'anticipated' and 'proposed' capacity and provide additional clarity for new investors. **Table 4** summarises the new project scoring criteria and weighting for categorising projects as used for the 2026 WEM ESOO forecast supply capacity modelling.

Table 5 compares the methodology and project categorisation in the 2026 and 2025 WEM ESOOs. All assessment was performed between March to April 2026, reflecting information available to AEMO at that time.

Table 4 New project evaluation methodology

Criteria name	Weighting	Status of new projects	Score (%)
Land approval	0.20	If the proponent has acquired all required approvals towards settlement or purchase or acquisition.	100
		If the proponent has submitted all applications towards purchase or acquisition but are still pending approval	50
		If the proponent has not yet applied for some or all the required approvals	0
Planning approvals	0.20	If the proponent has obtained an environmental approval or has exempted from it, and has obtained a development approval	100
		If the proponent has either of the following status: • either has obtained or has exempted from an environmental approval and has progressed for a development approval; or • has obtained a development approval and has progressed towards an environmental approval	75
		If the proponent has either of the following status: • either has obtained or has exempted from an environmental approval and yet to submit application for a development approval; or • has obtained a development approval and yet to submit application for an environmental approval; or • has submitted applications for both approvals	50
		If the proponent has applied for either an environmental approval or a development approval	25
		If the proponent has not yet applied for an environmental approval or a development approval	0
Western Power connection access	0.20	If a proponent's network access proposal contract has been signed with Western Power	100
		If a proponent's network access proposal has progressed sufficiently towards approval	50
		If a proponent has not yet submitted a network access proposal	0
Finance approval by proponent	0.20	If a proponent has the financing contract executed or has finalised board authority to proceed on an equity basis	100
		If the proponent's financing proposal is approved to a financier or has conditional approval to proceed on an equity basis	50
		If a proponent has submitted a financing proposal to a financier or submitted a request for approval to proceed on an equity basis	25
		If a proponent has not yet progressed towards financial approvals	0
Construction	0.20	If a proponent has completed construction	100
		If the proponent has commenced construction	50
		If the proponent has received a building permit but not yet commenced construction	25
		If the proponent has not yet applied for a building permit	0

Table 5 New project status evaluation methodology, 2026 and 2025 WEM ESOOs

	2026 WEM ESOO	2025 WEM ESOO
Assessment criteria	Five criteria: 1. Land approval 2. Planning approvals 3. Western Power Access 4. Finance approval towards FID 5. Construction approval	Three criteria: 1. Environmental approval 2. Western Power Access 3. Finance approval towards FID
New project categorisation	Three categories: 1. Committed capacity if project score is at least 80%. 2. Anticipated capacity if project score at least 60% but less than 80%. 3. Proposed capacity if project score is less than 60%.	Three categories: 1. Committed capacity if project score is at least 80%. 2. Probable capacity if project score more than 50% and less than 80%. 3. Proposed capacity if project score is less or equal to 50%.

The same project evaluation methodology is applied to new projects for inclusion in forecast of flexible capacity.

New projects that have assigned Flexible Capacity Credits for 2027-28 are considered as committed capacity. The timing of these projects are similarly monitored through project updates received from Western Power.

AEMO considers the responses received from FIR and 2026 EOI for Flexible CRC. The minimum eligibility criteria for Flexible Certified Reserve Capacity[33] then applies to all information received on projects to forecast for the fleet mix in this WEM ESOO.

A2.2 Changes in assigned Capacity Credits from 2026-27 to 2027-28

In addition to assessing new projects for their inclusion in the 2026 WEM ESOO, one of the key drivers for supply forecasts in this WEM ESOO is the changes in assigned Peak and Flexible Capacity Credits in the 2025 Reserve Capacity Cycle (for 2027-28). As Flexible Capacity Credits were assigned for the first time in the 2025 Reserve Capacity Cycle, only changes of assigned Peak Capacity Credits are considered in the following section.

A total of 6,375 MW of Capacity Credits were assigned to 93 Facilities for 2027-28. This is a 7% increase in assigned Capacity Credits for 2026-27 from 72 Facilities. Two key observations are:

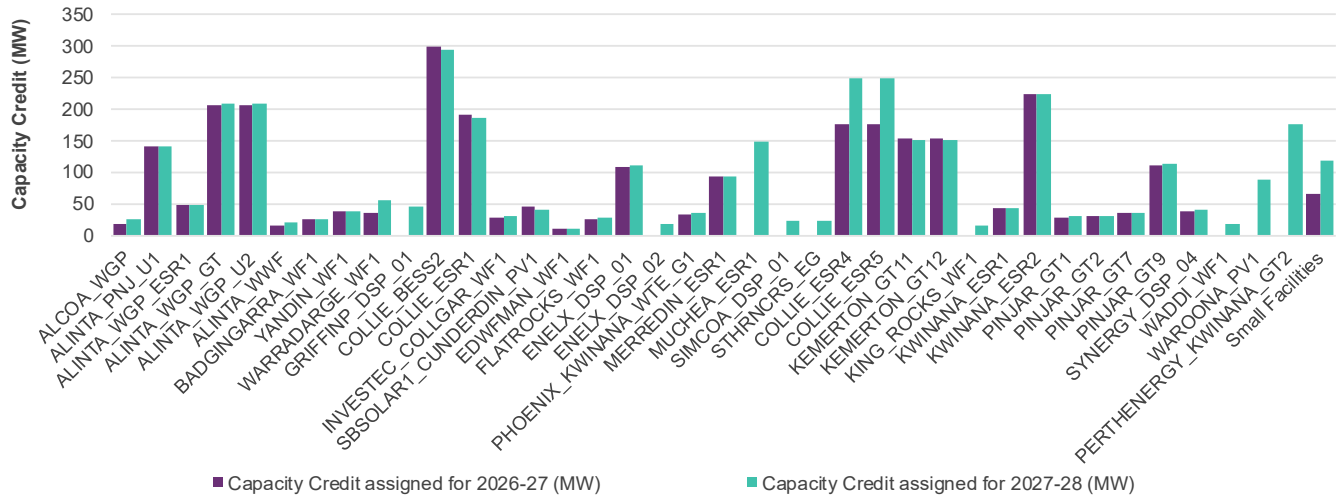
- Among the Capacity Credits assigned to new entrants in 2027-28, approximately 49% are Electric Storage Resource, followed by 32% of gas-fired Facilities while the remaining are a mix of thermal, renewable and Demand Side Program (DSP).
- This increase was partially offset by a 348 MW reduction of assigned Capacity Credits in 2027-28 from Facilities such as Collie Power Station ran by Synergy, Tesla Geraldton unit ran by Tesla Geraldton Pty Ltd, and DSP ran by Wesfarmers Kleenheat Gas Pty Ltd that did not seek Capacity Credits for that Capacity Year.

Figure 21 highlights the change in assigned Capacity Credits to Facilities between 2026-27 and 2027-28. Note a total of 21 Facilities that received the same Capacity Credits for these two Capacity Years are not presented in this figure. Four

[33] For detail on criteria for Flexible CRC see: https://www.aemo.com.au/energy-systems/electricity/wholesale-electricity-market-wem/wa-reserve-capacity-mechanism/certification-of-reserve-capacity/flexible-certified-reserve-capacity.

Facilities that did not receive Capacity Credits in 2027-28 are also excluded from this figure. **Appendix A3** includes a breakdown of changes to assigned Capacity Credits for the existing and committed Facilities.

Figure 21 Changes in assigned Peak Capacity Credits to Facilities, 2026-27 and 2027-28 (MW)



Note: "Small Facilities" refers to the sum of Capacity Credits for Facilities individually assigned less 10 MW of Capacity Credits in respective Capacity Years.

A3. Forecast supply capacity mix

A3.1 Forecast supply capacity mix in the 2026 WEM ESOO reliability modelling

Forecast supply capacity in the 2026 WEM ESOO refers to projected capacity mix which contributes to meeting forecast demand across the year, and includes the following groups of Facilities:

- grid-scale energy-producing Facilities (Non-Intermittent Generating Systems (NIGS), and Intermittent Generating Systems (IGS)), which include thermal and renewable generators,
- grid-scale energy storage (ESR) Facilities, which include battery and pumped-hydro storage, and
- Demand Side Programme (DSP) Facilities that reduce load or increase injection from non-dispatchable loads or embedded generation associated with a load.

Detail descriptions for each of these group of Facilities by their technology types are included in **Table 16** of Section 3.2.1 of the 2026 WEM ESOO report.

In this WEM ESOO, all projects that are modelled for forecasting supply capacity, are assessed and classified into existing, committed, anticipated, and proposed project categories, along with consideration of an estimated Relevant Level Method (RLM) uplift capacity. Description of these project categories and inclusion of them in modelling forecast supply capacity across the *Low*, *Expected*, and *High* scenarios are summarised in **Table 15** of Section 3.1 of the 2026 WEM ESOO report. Assessment of projects are as detailed in Appendix A2.1.

The forecast supply capacity from the existing and committed projects, included in the *Expected* scenario, is also used as a key input for the assessment of the Limb A, Limb B and Limb C of the Planning Criterion. The forecast capacity quantity for each individual Facility in these assessments is based on the Facilities' technology type.

A summary of the forecast supply capacity for each of these three Limbs of the Planning Criteria is provided below while the detail discussion on other inputs that set the requirements of each of these Limbs is included in Appendix A4:

- For the assessment of Limb A, AEMO used the forecast peak capacity mix (as defined in Section 3.1 of the WEM ESOO report) in the *Expected* scenario. This applies to all groups of Facilities, that is, NIGS, IGS, ESR and DSP. The forecast capacity of ESR accounts for forecast degradation over time and adjustments for ESR Duration Requirements applied to ESRs after 10th year from the first time they were assigned Capacity Credits. In addition to the information on project categories provided in Section 3.2 of the 2026 WEM ESOO report, each project category is further detailed in the following sections of this appendix.
- For the assessment of Limb B (using AEMO dispatch modelling), AEMO used the forecast supply capacity in the *Expected* scenario for all technology types of Facilities as used for the Limb A assessment, except for intermittent renewable generators (wind and PV). For these intermittent generators, a maximum injection capacity of each Facility is used, which is a Facility-specific, static value that determines the maximum quantity of power a Facility can inject at the connection point to meet SWIS operational demand. This capacity is derived from a Facility's nameplate capacity, maximum sent-out capacity and declared sent-out capacity. The maximum injection capacity of each wind or solar Facility is then multiplied by a modelled "normalised" availability trace (that is, a trace ranging from 0-1 MW,

representing intermittent wind or solar output for a 1 MW facility, developed by AEMO), which represents an interval-by-interval, location-specific profile of achievable generation output subject to prevailing weather conditions which is detailed in Appendix A4.2.

- For the assessment of Limb C, AEMO used the forecast flexible supply capacity in the *Expected* scenario for each eligible Facility. Forecast flexible supply capacity is detailed in Section 3.5 of the 2026 WEM ESOO report and methodology for assessing projects is in Appendix A2.

A3.2 Existing capacity with assigned Peak Capacity Credits

Existing projects, connected to the SWIS, are those that are operational at the time of developing the 2026 WEM ESOO.

An existing Facility that has been assigned Peak Capacity Credits through to 2027-28, is forecast to continue its service for the rest of this WEM ESOO outlook period (from 2028-29 to 2035-36). The forecast peak capacity is assumed to be equal to their respective assigned Peak Capacity Credits through to 2027-28, following which the forecast capacity accounts for degradation of ESRs overtime from 2028-29 to 2035-36, derating of ESRs based on the ESR Duration Requirements as applicable, and any possible reduction of Peak Capacity Credit caused by high historical Forced Outage rates. Additionally, capacity procured under 2024-26 or 2025-27 Peak Demand NCESS – Reliability Services are modelled as existing projects if operational at the time of the 2026 WEM ESOO modelling. For Facilities that are assumed to retire or otherwise be unavailable, the forecast capacity reverts to zero from the retirement or assumed unavailability date until the end of the outlook period.

Table 6 presents the existing Facilities, excluding DSPs, that are considered in the 2026 WEM ESOO *Low*, *Expected* and *High* scenarios with their respective assigned Peak Capacity Credits in the past two Reserve Capacity Cycles. It also shows the maximum quantity of power that an individual Facility can inject to the SWIS, derived from its nameplate capacity, maximum sent-out capacity and declared sent out capacity. Similarly, **Table 7** presents the existing DSPs.

Table 6 Capacity of existing Facilities, excluding DSPs (MW)

Market Participant	Existing Facility	Peak Capacity Credits 2026-27	Peak Capacity Credits 2027-28	Maximum quantity of power injectable to the SWIS	Retirement or assumed unavailability in 2026 WEM ESOO modelling
Alcoa of Australia	ALCOA_WGP	18.880	26.000	26.000	
Alinta Sales Pty Ltd	ALINTA_PNJ_U1	141.164	142.447	143.270	
	ALINTA_PNJ_U2	142.447	142.447	143.270	
	ALINTA_WGP_GT	207.076	208.000	212.000	
	ALINTA_WGP_U2	207.076	208.000	212.000	
	ALINTA_WWF	16.859	21.011	89.100	
	BADGINGARRA_WF1	26.098	26.117	149.250	
	YANDIN_WF1	37.504	37.500	211.680	
Amanda Energy Pty Ltd	BIOGAS01	0.425	0.425	2.000	
BEI WWF Pty Ltd ATF WWF Trust	WARRADARGE_WF1	34.902	56.388	183.600	
Blair Fox Pty Ltd AFT Blair Fox Trust	BLAIRFOX_KARAKIN_WF1	0.093	0.045	5.000	

Market Participant	Existing Facility	Peak Capacity Credits 2026-27	Peak Capacity Credits 2027-28	Maximum quantity of power injectable to the SWIS	Retirement or assumed unavailability in 2026 WEM ESOO modelling
Bluewaters Power 1 Pty Ltd	BW1_BLUEWATERS_G2	217.000	217.000	217.000	1 October 2031
Bluewaters Power 2 Pty Ltd	BW2_BLUEWATERS_G1	217.000	217.000	217.000	1 October 2031
Collie Battery Pty Ltd ATF Collie Battery Trust	COLLIE_BESS2	300.000	293.560	300.000	
	COLLIE_ESR1	192.425	185.975	200.000	
Collgar Wind Farm	INVESTEC_COLLGAR_WF1	27.964	29.982	222.000	
Cunderdin Development Pty Ltd	SBSOLAR1_CUNDERDIN_PV1	47.070	40.794	155.000	
Denmark Community Windfarm Ltd	DCWL_DENMARK_WF1	0.571	0.566	1.440	
EDWF Manager Pty Ltd	EDWFMAN_WF1	11.356	10.409	99.200	
Enel Energy Australia Pty Ltd	FLATROCKS_WF1	25.341	27.432	73.400	
Kwinana WTE Project Co Pty Ltd	PHOENIX_KWINANA_WTE_G1	33.909	35.684	43.000	
Landfill Gas & Power Pty Ltd	RED_HILL	2.361	2.274	3.640	
	TAMALA_PARK	3.584	3.242	5.770	
MEG HP1 Pty Ltd	NORTHAM_SF_PV1	0.839	0.560	9.800	
Merredin Energy	NAMKKN_MERR_SG1	82.000	82.000	82.000	
Merredin Project Company Pty Ltd ATF The Merredin	MERSOLAR_PV1	7.610	5.561	100.000	
Metro Power Company Pty Ltd	AMBRISOLAR_PV1	0.739	0.710	0.960	
Mt.Barker Power Company Ltd	SKYFRM_MTBARKER_WF1	0.947	0.924	2.430	
Mumbida Wind Farm Pty Ltd	MWF_MUMBIDA_WF1	7.561	7.996	55.000	
NewGen Neerabup Partnership	NEWGEN_NEERABUP_GT1	330.600	330.600	330.600	
NewGen Power Kwinana Pty Ltd	NEWGEN_KWINANA_CCG1	327.800	327.800	327.800	
Perth Energy Pty Ltd	ROCKINGHAM	1.012	0.999	4.000	
	SOUTH_CARDUP	1.707	1.600	4.158	
PRD SWIS OPS PTY LTD	PRDSO_WALPOLE_HG1	1.500	1.500	1.500	
SRV AGWF Pty Ltd as Trustee for AGWF Trust	ALBANY_WF1	7.736	7.986	21.600	
	GRASMERE_WF1	5.434	5.552	13.800	
SRV GRSF pty Ltd as Trustee for GRSF Trust	GREENOUGH_RIVER_PV1	4.394	3.403	40.000	
Synergy	BREMER_BAY_WF1	0.246	0.227	1.880	
	COCKBURN_CCG1	240.000	240.000	240.000	
	COLLIE_G1	317.200	0.000	317.200	1 October 2027
	COLLIE_ESR4	176.010	250.000	250.000	

Market Participant	Existing Facility	Peak Capacity Credits 2026-27	Peak Capacity Credits 2027-28	Maximum quantity of power injectable to the SWIS	Retirement or assumed unavailability in 2026 WEM ESOO modelling
	COLLIE_ESR5	175.347	250.000	250.000	
	KALBARRI_WF1	0.114	0.096	1.600	
	KEMERTON_GT11	155.000	151.919	155.000	
	KEMERTON_GT12	155.000	151.919	155.000	
	KWINANA_ESR1	43.500	42.750	100.000	
	KWINANA_ESR2	225.000	224.045	225.000	
	KWINANA_GT2	106.763	106.763	106.763	
	KWINANA_GT3	106.804	106.804	106.804	
	MUJA_G7	211.000	211.000	211.000	1 October 2029
	MUJA_G8	211.000	211.000	211.000	1 October 2029
	PINJAR_GT1	29.070	32.000	33.400	1 October 2031
	PINJAR_GT2	31.600	29.720	31.600	1 October 2031
	PINJAR_GT3	37.000	37.000	37.000	1 October 2031
	PINJAR_GT4	35.950	35.950	35.950	1 October 2031
	PINJAR_GT5	37.000	37.000	37.000	1 October 2031
	PINJAR_GT7	35.750	35.166	38.300	1 October 2031
	PINJAR_GT9	110.500	114.314	116.300	1 October 2032
	PINJAR_GT10	110.500	110.500	110.500	1 October 2034
	PINJAR_GT11	123.350	123.350	123.350	
Tesla Corporation Management	TESLA_PICTON_G1	0.000	9.999	9.999	
Tesla Geraldton Pty Ltd	TESLA_GERALDTON_G1	9.999	0.000	9.999	1 October 2027
Tesla Kemerton Pty Ltd	TESLA_KEMERTON_G1	9.205	9.999	9.999	
Tesla Northam Pty Ltd	TESLA_NORTHAM_G1	0.000	0.000	9.900	1 October 2026
Tronox Management Pty Ltd	TIWEST_COG1	36.000	36.000	36.000	31 December 2034
Waste Gas Resources Pty Ltd	HENDERSON_RENEWABLE_IG1	1.476	1.474	3.192	
Western Energy Pty Ltd	PERTHENERGY_KWINANA_GT1	109.000	109.000	109.000	

Table 7 Existing DSP and availability[A] (MW)

Market Participant	DSP Name	Capacity Credits 2026-27	Capacity Credits 2027-28	Maximum quantity of power injectable to the SWIS
Bluewaters Power 1 Pty Ltd	GRIFFINP_DSP_01	0.000	45.010	45.010
Synergy	SYNERGY_DSP_04	38.123	42.000	42.000
Wesfarmers Kleenheat Gas Pty Ltd[B]	PREMPWR_DSP_02	20.994	-	20.994

A. DSPs must be available to provide at least 50 hours of Reserve Capacity during 2026-27 and at least 23.75 hours during 2027-28. From 2028-29 and onward, DSPs must be available to provide at least 28.625 hours of Reserve Capacity during a Capacity year. This availability must cover no less than 12 hours per Business Day between 08:00 and 20:00.

B. AEMO received notification from Wesfarmers Kleenheat Gas Pty Ltd that they no longer plan to apply for Certified Reserve Capacity for their listed DSP Facility from 2025 Reserve Capacity Cycle onwards. For detail see https://www.aemo.com.au/energy-systems/electricity/wholesale-electricity-market-wem/wa-reserve-capacity-mechanism/notifications-of-facilities-ceasing-operation-in-the-wem.

A3.3 Forecast committed capacity

Projects that have been assigned Capacity Credits for the first time in 2026-27 or 2027-28 were modelled to be in operation from the respective Capacity Years as committed capacity. Any project that has been assigned Capacity Credits in earlier Capacity Years but are not online at the time of this WEM ESOO modelling, were also considered as committed capacity for the year when it is forecast to be online, based on assessment of information received from progress reports, Western Power and AEMO market research. Additionally, capacity procured under 2025-27 Peak Demand NCESS – Reliability Services are modelled as committed projects, if their services had not been activated at the time of this WEM ESOO development. Projects that scored above 80% in the new project status evaluation (see Appendix A2.1) were also modelled as committed capacity. The start dates of these projects were assessed based on information received through the 2025 Long Term PASA FIRs from Market Participants, project updates from Western Power as part of this WEM ESOO development process, and submissions for 2026 EOIs for CRC. These committed projects are summarised in **Table 8.**

Table 8 Forecast supply capacity[A] from committed projects, by technology type[B] (MW)

Generation and storage technology type	Forecast supply capacity in 2026-27	Forecast supply capacity in 2027-28	Forecast supply capacity in 2028-29	Forecast supply capacity in 2029-30	Forecast supply capacity in 2030-31	Forecast supply capacity in 2035-36	Maximum quantity of power injectable to the SWIS
Wind	0	35	105	190	350	346	1,618
Gas	85	284	285	285	285	285	299
Solar	0	2	9	9	9	9	106
Waste-to-energy	25	0	25	25	25	25	29
ESR[C]	143	324	399	392	385	350	488
DSP	9	41	41	41	41	41	41
Distillate	0	25	25	25	25	25	29
Total without estimated Relevant Level Method (RLM) uplift capacity	262	711	889	967	1,119	1,081	2,609
Estimated RLM uplift capacity[D]	-	-	240	240	240	240	-
Total with estimated RLM uplift capacity	262	711	1,129	1,207	1,359	1,321	-

A. All figures have been rounded to the nearest MW. Consequently, totals may have a 1 MW difference due to rounding.
B. The components of hybrid Facilities are split by their respective technology types.
C. ESR capacity is modelled to degrade over the outlook period.
D. RLM uplift capacity is as defined and detailed in Section 3.4.2 of the 2026 WEM ESOO report.

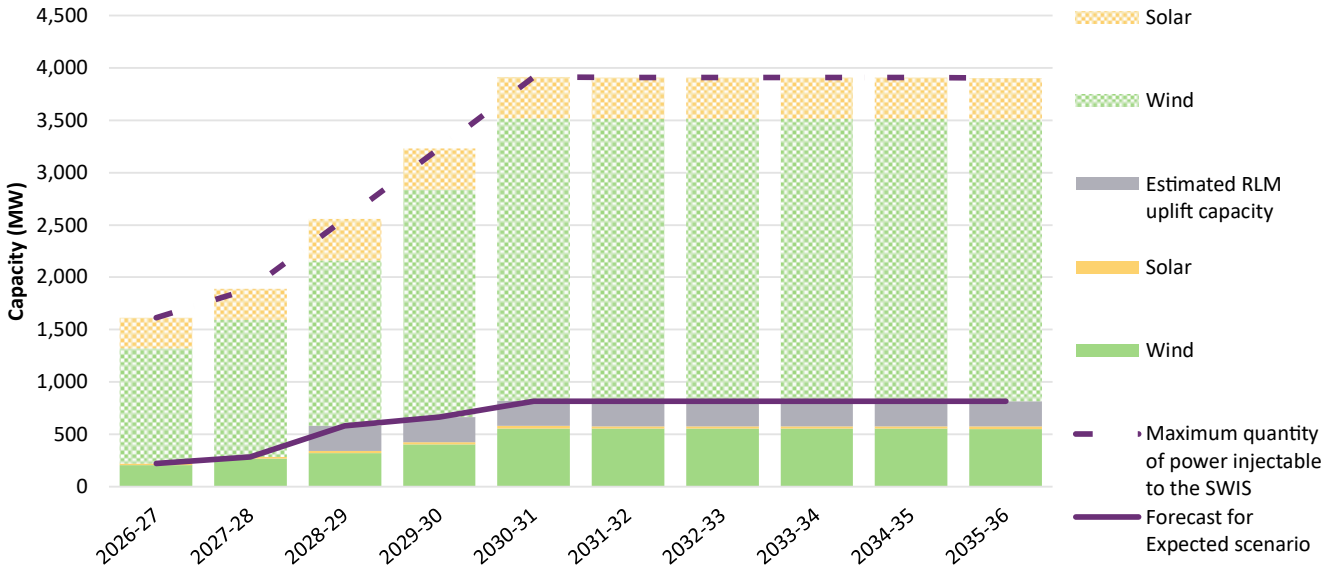
Forecast supply capacity, especially from intermittent generators, can be noticeably lower than their maximum quantity of power injectable to the SWIS

The forecast supply capacity of some Facilities can be noticeably lower than their maximum quantity of power injectable to the SWIS. This is reflected from the Facilities with assigned Capacity Credits in the 2024 and 2025 Reserve Capacity Cycles (existing and committed Facilities presented in **Table 6** and **Table 8** above) in which assigned Capacity Credits are lower than the Facilities' corresponding maximum quantity of power injectable to the SWIS.

Facilities may not be assigned Capacity Credits for reasons including network limitations and their prioritisation in the NAQ model, as discussed in Section 3.6.2 of the 2026 WEM ESOO report.

Figure 22 highlights the solar and wind generators that have the highest difference between their forecast capacity and corresponding maximum quantity of power injectable to the SWIS compared to any other group of Facilities. It shows the significant difference across all the Capacity Years from 2026-27 to 2035-36. This WEM ESOO forecasts an estimated RLM uplift capacity of 240 MW applicable from the 2026 Reserve Capacity Cycle onwards. The impact of this estimated RLM uplift capacity on the total forecast supply capacity is detailed in Section 3.4.2 of the 2026 WEM ESOO, and additional information is available on the AEMO website[34].

Figure 22 Difference between forecast capacity and maximum quantity of power injectable to the SWIS, wind and solar generators, *Expected* scenario, 2026-27 to 2035-36 (MW)



A3.4 Forecast anticipated capacity

Projects that scored between 60% and 80% in the new project status evaluation described in Appendix A2.1 were categorised as anticipated capacity and modelled in the 2026 WEM ESOO *High* scenario to commence operation from 2028-29 onwards (see **Table 9**).

Table 9 Forecast supply capacity from anticipated projects, by technology type, 2026-27 to 2035-36 (MW)

Generation and storage technology type	Forecast supply capacity in 2026-27	Forecast supply capacity in 2027-28	Forecast supply capacity in 2028-29	Forecast supply capacity in 2029-30	Forecast supply capacity in 2030-31	Forecast supply capacity in 2035-36	Maximum quantity of power injectable to the SWIS
Solar	0	0	0	2	2	7	129
ESR	0	0	622	673	659	590	780
Total	0	0	622	675	661	596	909

Note: All figures have been rounded to the nearest MW. Consequently, totals may have a 1 MW difference due to rounding.

[34] See https://www.aemo.com.au/energy-systems/electricity/wholesale-electricity-market-wem/wa-reserve-capacity-mechanism/certification-of-reserve-capacity/relevant-level-methodology.

A4. Reliability assessment framework, methodology and results

A4.1 Reliability assessment framework

Reliability of supply, or supply adequacy, is the ability of a power system to meet electricity demand at all times, with consideration for a prescribed tolerance.

Reliability of supply is measured in units of unmet demand (MW) or unserved energy (USE, MWh) [35]. In this report, reliability of supply is measured as annual USE relative to annual energy consumption, and compared against the 0.0002% reliability standard set by Limb B of the Planning Criterion (clause 4.5.9(b) of the ESM Rules).

The assessment of SWIS reliability is based on forecasts of consumption and demand, forecast of supply, and – for certain components of the assessment – forecasts of network congestion.

Consumption and demand forecasts include annual energy consumption (MWh), annual peak demand (MW), and demand (MW) in all half-hourly intervals of a year (time-sequential demand profile, or trace). These forecasts consider various types of loads, growth drivers and the impact of DER. Details on consumption and demand forecasts are provided in Section 2 and Appendix A1.

Supply forecasts provide an outlook on the capacity of existing and new-entrant generation, storage and demand-side response Facilities. These forecasts also assess the Facilities' ability to inject or withdraw power under various operating conditions (such as outages, or availability of wind and solar resource) to meet SWIS operational demand, subject to a set of operating constraints. Details on supply forecasts are in Section 3, Appendix A2 and Appendix A3.

Network congestion forecasts include an outlook on the capability of the SWIS transmission network to transfer generation output to meet demand at the sub-regional level. Details on network forecasts are in Section 6 and Appendix A4.3.

The SWIS reliability assessment is performed against a set of metrics and requirements, presented in **Table 10**.

Table 10 The requirements and metrics of the SWIS reliability assessment framework

Category	Component	Clause	Comments
Planning Criterion	Limb A requirement	4.5.9(a), 4.5.10(b), 4.5.2A(b)	Requires that the SWIS has sufficient supply capacity to meet the forecast, system-wide, annual unscheduled operational peak demand (MW), plus: • a reserve margin • an allowance for the Regulation Raise requirement • an allowance for Intermittent Loads • the ESR Duration Requirement Uplift. Assessed for the 10% probability of exceedance (POE) *Expected* scenario, across all years of the outlook period.

[35] In the case of half-hourly intervals, unserved energy is the sum of unserved demand divided by two.

Category	Component	Clause	Comments
	Limb B requirement	4.5.9(b), 4.5.10(b)	Requires that the SWIS has sufficient supply capacity to limit USE to no more than 0.0002% of annual energy consumption. This means that SWIS supply capacity must meet at least 99.9998% of the forecast, system-wide, annual unscheduled energy consumption (MWh). This assessment uses dispatch modelling of the supply capacity (without network constraint equations) to meet demand in each interval of a year, and identifies any intervals when demand is unable to be served. The annual sum of unmet demand (MW), divided by two due to half-hourly intervals, is USE (MWh). Assessed for the 10% POE *Expected* scenario, across all years of the outlook period.
	Limb C requirement	4.5.9(c), 4.5.10(bA)	Requires that the SWIS has sufficient supply capacity to be able to ramp up quickly enough to meet the highest forecast growth of demand over four consecutive hours (MW) plus a reserve margin. Assessed for the 10% POE *Expected* and the 50% POE *Expected* scenario, across all years of the outlook period.
RCTs	Peak RCT	4.5.10(b)	Informed by Limb A, Limb B and Limb C assessments, the Peak RCT is set by the highest of the requirements determined by Limb A, Limb B, and Limb C. Indicates how much peak capacity is required for maintaining reliability at the SWIS-wide level, without consideration for network constraints. Assessed across all years of the outlook period.
	Flexible RCT	4.5.10(bA)	Set as the higher of the 10% POE *Expected* scenario and the 50% POE *Expected* scenario assessments of the Limb C requirement. Indicates how much flexible capacity is required for maintaining reliability at the SWIS-wide level without consideration for network constraints. Assessed across all years of the outlook period.
Supply-demand balance	Peak RCT and Peak Capacity; Flexible RCT and Flexible Capacity	4.5.10(aA)	This assessment compares the amount of peak capacity and flexible capacity (assessed with regard for the impact of network congestion under the NAQ framework) against respective targets, and identifies supply capacity shortfalls, if any. Assessed across all years of the outlook period.
Capability Class assessment	Limb B USE and peak capacity	4.5.12(i)	The purpose of the Capability Class assessment is to determine how much supply capacity is required from Capability Class 1 and Capability Class 3 Facilities. Assessed for the third year of the outlook period if USE modelled under Limb B of the Planning Criterion exceeds the 0.0002% SWIS reliability standard.
Availability Curves	Demand profiles and margin	4.5.10(e)	Curves representing demand in all Trading Intervals of a year (ordered from highest to lowest), increased by a margin. Availability Curves are developed for the second and third year of the outlook period, and represent the required minimum amount of peak capacity in each of the Trading Intervals. Assessed across all years of the outlook period.
Sub-regional assessment	Regional capacity shortfalls	4.5.10(c)	Assessed based on the same dispatch modelling framework as used in Limb B assessment, but inclusive of network constraints to identify USE in sub-regions of the SWIS, and at the Regional Reference Node (RNN). Assessed across all years of the outlook period.
	Alleviation options	4.5.10(d)	Identification of transmission, generation, storage or demand-side capacity augmentation options to alleviate capacity shortfalls due to network constraints.
ESR requirements	ESR Duration Requirement	4.5.12(d)	The ESR Duration Requirement is the number of consecutive Trading Intervals over which ESR is required to be available for dispatch. Assessed for the third Capacity Year of the study horizon, based on a reviewed method for Availability Gap Duration (ADG) calculation, as per the ESM Amendment (Tranche 8) Amending Rules.
	ESR Duration Requirement (ESRDR) Uplift	4.5.12(e)	The ESRDR Uplift accounts for the need for additional Reserve Capacity due to an increase to the ESR Duration Requirement and the existing protection mechanism in the ESM Rules for ESR previously certified on the basis of a shorter ESR Duration Requirement. Assessed for the third Capacity Year of the study horizon, informs the Limb A assessment.
	Mid Peak ESROI	4.11.3A(a), 4.11.3A(b)	MP ESROI is the timing of the reference Trading Interval which, in conjunction with the ESR Duration Requirement, sets the Peak ESR Obligation Duration, that is the timing of consecutive Trading Intervals over which ESR is required to be available for dispatch. Determined in line with a relevant WEM Procedure.

Category	Component	Clause	Comments
	Flexible ESROI	4.11.3A(aA), 4.11.3A(b)	Represents the timing of the last Trading Interval which, in conjunction with the Flexible Capacity Obligation Duration[A], sets the timing of consecutive Trading Intervals over which Flexible ESR is required to be available for dispatch.
DSP requirements	Indicative DSP Dispatch Threshold	4.5.12(f)	The level of unscheduled operational demand above which capacity from DSPs is expected to be required. Assessed for the third Capacity Year of the study horizon, based on the 50% POE *Expected* demand scenario.
	Peak DSP Dispatch Requirement	4.5.12(g)	The annual number of Trading Intervals exceeding the Indicative DSP Dispatch Threshold, that is the annual number of Trading Intervals over which DSP providing peak capacity is expected to be required. Assessed for the third Capacity Year of the study horizon.
	Flexible DSP Dispatch Requirement	4.5.12(h)	the annual number of Trading Intervals exceeding the Indicative DSP Dispatch Threshold, that is the annual number of Trading Intervals over which DSP with Flexible Capacity Credits is expected to be required in addition to its Peak DSP Dispatch Requirement. Assessed for the third Capacity Year of the study horizon.

A. As per the ESM Rules, the Flexible Capacity Obligation Duration is eight Trading Intervals.

A4.2 Approach to determining the amount of capacity required to satisfy Limb A

The Limb A requirement is calculated based on the values projected for each of the Limb A building blocks, summarised in **Table 11**. Detail on how the 10% POE peak demand is adjusted with impacts of the VPP, reserve margin, Regulation Raise, and ESRDR Uplift components is provided below.

Table 11 Summary of building blocks of Limb A

Building block of Limb A	Description
10% POE peak demand	Forecast annual operational sent-out peak demand for the 10% POE *Expected* scenario, accounting for the impacts of the VPP. Refer to Section 2 for detail.
Intermittent loads	Estimate of the capacity required to cover the forecast cumulative requirements of Intermittent Loads, which are excluded from the 10% POE peak demand forecast. Estimation of this capacity is equivalent to the total over all expected Intermittent Loads of their forecast maximum possible Intermittent Load levels multiplied by: • The ratio of: - The Peak Reserve Capacity Target for each Capacity Year, and - The forecast 10% POE peak demand for the relevant Capacity Year • Minus one.
Reserve margin	Determined as the greater of: • Forced Outage allowance, determined as peak demand multiplied by the proportion of Capacity Credits expected to be unavailable at the time of peak demand due to Forced Outages, and • the largest contingency relating to loss of supply (related to any Facility, including a Network) at the time of peak demand. This considers the historical Forced Outages reflecting the performance of the SWIS generation fleet and the need for AEMO to maintain real-time reserves for Contingency Reserve Raise. For the purposes of this WEM ESOO, AEMO has calculated the largest risk as equivalent to the loss of the two largest generating units plus 70% of the third largest unit for each Capacity Year over the 10-year outlook period.
Regulation Raise	Accounts for the requirement of Regulation Raise (a service that increases grid frequency by adjusting generation or demand in response to variation) and is escalated to account for the impact of increasing deployment of intermittent generators (wind, large-scale solar, distributed PV) on the forecast requirement of Regulation Raise.
ESR Duration Requirement Uplift	The ESR Duration Requirement Uplift quantifies the overallocation of Capacity Credits assigned to ESR in the historical Reserve Capacity Cycles that have shorter duration than required by the ESR Duration Requirement (in accordance with grandfathering provisions).

Between 2026-27 to 2028-29, the Limb A requirement is lower compared to the requirements set in the 2025 WEM ESOO, primarily driven by the lower reserve margin and Regulation Raise requirements determined in the 2026 WEM ESOO.

From 2029-30 to 2034-35, Limb A requirements are higher compared to the requirements set in the 2025 WEM ESOO, primarily driven by higher ESR Duration Requirement Uplift and higher peak demand forecasts in the 2026 WEM ESOO.

Table 12 compares building blocks of Limb A requirements in the 2026 WEM ESOO with those blocks in the 2025 WEM ESOO. Sections A4.2.1 to A4.2.4 further detail the building blocks of the 2026 WEM ESOO Limb A requirements.

Table 12 Comparison of Limb A building blocks, 2026 and 2025 WEM ESOO, 2026-27 to 2034-35 (MW)

Capacity Year	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34	2034-35
10% POE peak demand	-74	22	-1	45	82	120	88	79	89
Intermittent loads	-4	-4	-4	-4	-4	-4	-4	-4	-4
Reserve margin	-96	-94	-97	-100	-103	-105	-108	-111	-113
Regulation raise	-34	-41	-40	-29	-16	-20	-22	-22	-22
ESR Duration Requirement Uplift	-	0	136	136	136	136	136	136	136
Limb A requirement (net change)	**-208**	**-117**	**-7**	**47**	**93**	**126**	**90**	**79**	**86**

Note: A positive value indicates 2026 WEM ESOO forecasts are higher than the corresponding 2025 WEM ESOO values, while a negative value indicates the opposite.

A4.2.1 Approach to determining the peak demand building block of Limb A

The 10% POE *Expected* annual peak demand forecasts used in the Limb A assessment is prepared in line with AEMO's *Electricity Demand Forecasting Methodology*, and includes the impact of VPP (coordinated residential batteries), modelled for the first time in the 2026 WEM ESOO. For details, see Section 2.5 of the 2026 WEM ESOO report and Appendix A1.13.

A4.2.2 Approach to determining the reserve margin building block of Limb A

The reserve margin of Limb A is determined as the greater of the Forced Outage allowance, and the largest supply contingency at peak demand as defined in **Table 12** above.

For the determination of the reserve margin in this WEM ESOO, AEMO analysed historical Forced Outage data as well as identified the need to maintain real-time reserves for Contingency Reserve Raise. The reserve margin is determined to be equivalent to the loss of the two largest generating units plus 70% of the third largest unit for each Capacity Year over this 10-year outlook period. This is detailed in Section 3.7 of the 2026 WEM ESOO report.

Forced Outage allowance methodology

AEMO assessed the Forced Outage allowance via the following method:

- AEMO evaluated the anticipated installed capacity (AIC) and Forced Outage rates[36] of existing Facilities following the CRC assessment methodology[37].

[36] The historical 36-month Forced Outage analysis is based on data as of 26 April 2026.

[37] The WEM Procedure: Certification of Reserve Capacity. See https://www.aemo.com.au/energy-systems/electricity/wholesale-electricity-market-wem/procedures-policies-and-guides/procedures.

- The expected unavailable capacity was calculated by multiplying AIC with the Forced Outage rate for each Facility[38].
- The proportion of Capacity Credits expected to be unavailable at the time of peak demand due to Forced Outages is the sum of expected unavailable capacity divided by the total AIC.
- The Forced Outage allowance is the forecast 10% POE peak demand multiplied by the proportion of Capacity Credits expected to be unavailable at the time of peak demand due to Forced Outages.

Using this approach, the expected unavailable capacity ranges from 215 MW in 2026-27 to 275 MW by 2035-36.

Largest contingency assessment methodology

The largest contingency assessment involves factors such as fuel disruptions, project delays and retirements, Forced Outage frequency during peak conditions, unplanned generation exits, degradation of grid-scale energy storage, and network element failures. The methodology used in the 2026 WEM ESOO has been expanded from previous WEM ESOOs to consider broader criteria to maintain reliability of the SWIS. The high-level steps are:

- analysis of the Forced Outages of operational Facilities determining the Forced Outage frequency of the SWIS power system in peak intervals during the previous six Hot Seasons (2020-21 to 2025-26),
- detailed assessment of historical market data on the Contingency Reserve Raise requirement that indicates reserve requirements during extreme peak conditions are typically below 70% of the largest contingency,
- alignment with Medium Term PASA and Short Term PASA operational reserve requirements supporting maintenance of the SWIS in a normal operating state without requiring AEMO intervention to maintain power system security and reliability, and
- alignment to clause 4.5.9(a)(i) of the ESM Rules.

All six Hot Seasons experienced instances where the total capacity on Forced Outage exceeded the magnitude of the two largest generating units. This occurred during 4% and 2% of the peak intervals in the 2023-24 and 2024-25 Hot Seasons, respectively (down from 14% in the 2022-23 Hot Season). Incidents in the 2022-23 Hot Season included up to three units.

Based on AEMO's assessment, there is significant risk that the SWIS may need to operate with Forced Outages exceeding the capacity of the two largest generating units based on the analysis of historical Forced Outages. An additional loss equivalent to 70% of the third largest Facility is considered to manage real-time operational contingencies during peak demand conditions. The results are detailed in Section 3.7 of the 2026 WEM ESOO report.

Following AEMO's assessment using the Forced Outage allowance approach and the largest contingency approach, the reserve margin building block of Limb A is set by the largest contingency as the greater of the two. This reserve margin would ensure Power System Security & Reliability (PSSR) can be maintained during any Outage equivalent of the two largest generators on Outage while maintaining a real-time reserve equivalent to the 70% of the third largest unit.

The reserve margin is calculated based on the assigned Capacity Credits to SWIS generators and ESRs (accounting for any forecast degradation of ESRs) as detailed in Section 3.7 of the 2026 WEM ESOO report. The reserve margin reduces from 880 MW in 2026-27 to 864 MW in 2027-28 following the announced retirement of Collie Power Station and further decreases through to the end of this outlook, reaching 842 MW in 2034-35 due to forecast degradation of ESR capacity.

[38] Excluding Forced Outages of Facilities, to which clause 4.11.1A of the WEM Rules are expected to apply.

Compared to the 2025 WEM ESOO, across the 2026 WEM ESOO outlook period there is a 103 MW average annual reduction in the reserve margin component as a result of the revised assessment of this margin.

A4.2.3 Approach to determining the Regulation Raise building block of Limb A

AEMO estimated the minimum Regulation Raise requirement using the Frequency Co-optimised Essential System Services (FCESS) framework, ensuring frequency is maintained within the Normal Operating Frequency Band. Forecast quantities account for the increasing penetration of intermittent generation and distributed PV, building on the 2025 WEM ESOO approach.

Regulation Raise is formulated as the existing Regulation Raise requirement escalated by:

- 3 % of nameplate capacity of new wind generation in North Country, Mid West, Central Midlands, and Neerabup nodes,
- 1.8% of nameplate capacity of new wind generation at other nodes,
- 5% of nameplate capacity of new grid-scale solar generation, and
- 4% of new distributed PV nameplate capacity.

For the 2026 WEM ESOO, the distributed PV component is presented net of DESS capacity. This reflects the growing uptake of behind-the-meter storage and its impact on absorbing surplus distributed PV output that would otherwise increase the volatility of operational demand and result in a higher Regulation Raise requirement. The Regulation Raise building block is calculated for each Capacity Year and detailed in Section 3.8 of the 2026 WEM ESOO.

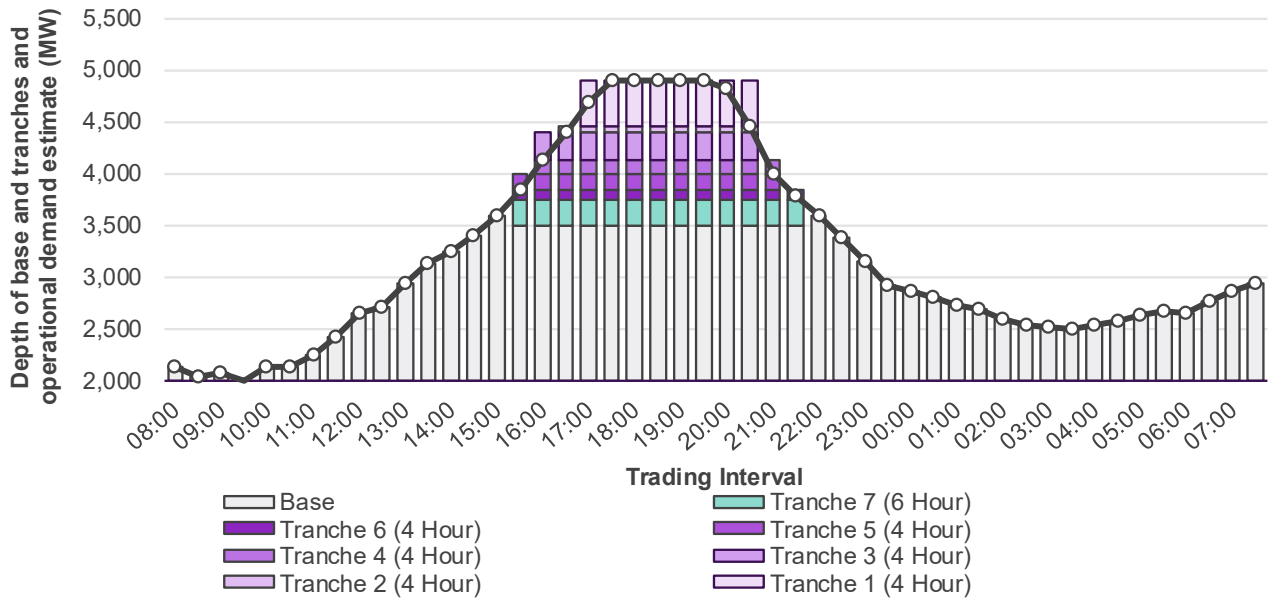
A4.2.4 Approach to determining the Electric Storage Resource Duration Requirement Uplift building block of Limb A

The ESRDR Uplift, determined in accordance with Appendix 11 of the ESM Rules, represents an additional building block of Limb A introduced as part of ESM amendment (Tranche 8) Rules in June 2025.

AEMO applied the methodology prescribed under Appendix 11 of the ESM Rules to the peak demand day in each weather reference year for the 2028-29 Capacity Year. **Figure 23** shows an example of the calculated dispatch of ESR in a single weather reference year.

In this 2026 WEM ESOO, the methodology now requires separate treatment of storage of different durations in accordance with Appendix 11 of the ESM Rules.

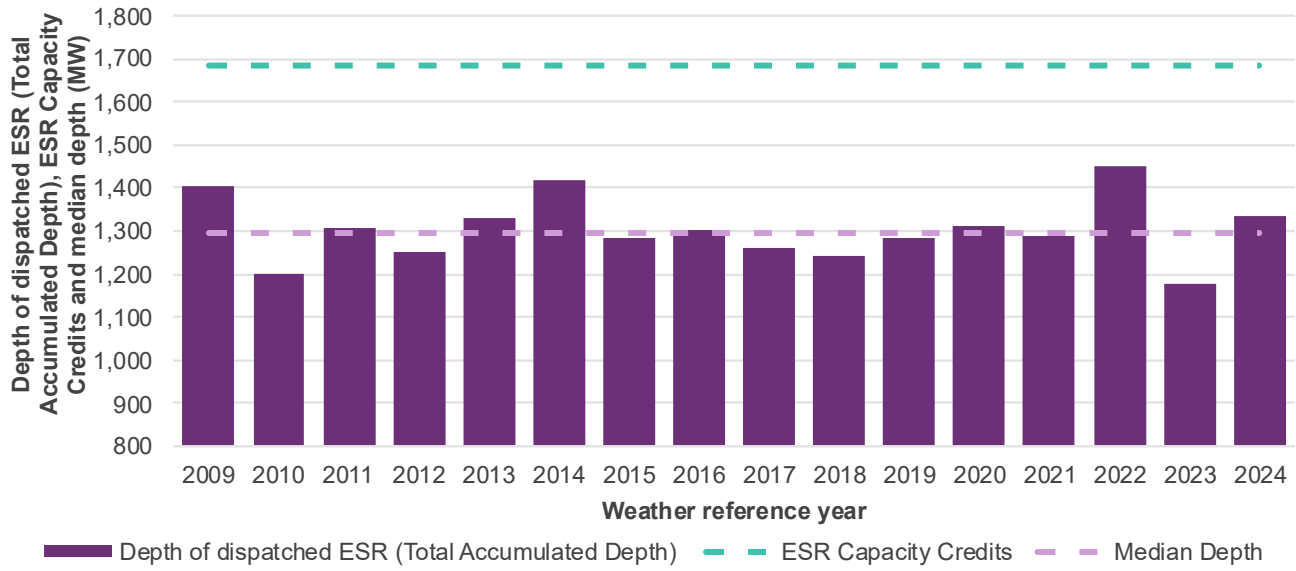
Figure 23 Depth of tranches and operational demand estimate for ESR Duration Requirement Uplift calculation based on the 2009-10 weather reference year (MW)



The depth of the dispatched tranches (Total Accumulated Depth) is then compared to the Capacity Credits assigned to those same Facilities. The median value across assessed weather reference years is determined as the ESRDR Uplift for the 2028-29 Capacity Year.

Figure 24 shows the depth for each assessed weather reference year alongside the assigned Capacity Credits for ESR, and the median difference between the depth and the Capacity Credits sets the uplift quantity. For the 2028-29 Capacity Year, the ESRDR Uplift has been determined as 392 MW.

Figure 24 Depth of dispatched ESR (Total Accumulated Depth), ESR Capacity Credits and median depth for each weather reference year for ESR Duration Requirement Uplift calculation (MW)



A4.3 Approach to determining the amount of capacity required to satisfy Limb B

Dispatch modelling methodology

The Limb B assessment stipulates that there should be sufficient available capacity to limit expected energy shortfalls to 0.0002% of annual energy consumption (MWh). This assessment uses dispatch modelling of the supply capacity to meet demand, and identifies any intervals when demand is unable to be served. The annual sum of unmet demand (MW), divided by two due to half-hourly intervals, is annual unserved energy (USE, MWh).

AEMO has brought the electricity market dispatch modelling in-house for the first time this WEM ESOO. AEMO uses PLEXOS to model co-optimised dispatch of energy and ESS using a time-sequential, half hour interval-by-interval modelling approach.

The dispatch modelling underpinning the Limb B assessment was undertaken using:

- Demand and renewable traces:
  - AEMO developed half-hourly profiles (traces) of unscheduled operational demand and available renewable generation. Demand traces incorporate distributed PV forecasts and account for the impact of DER, including the impact of the VPP (see Appendix A1.13). Profiles of intermittent renewables availability reflect location-specific weather conditions for existing and committed intermittent Facilities (wind and solar).
- Forecasts of supply capacity:
  - AEMO's forecast of supply capacity used in dispatch modelling considered generation (thermal and renewable), storage and DSPs which are either existing (assigned Capacity Credits or NCESS contracts) or are considered by AEMO to be committed as detailed in Appendix A2. Forecast capacity was assessed as:
    - for thermal Facilities – assigned Capacity Credits or procured Peak Demand NCESS – Reliability Services or forecast capacity,
    - DSP – assigned Capacity Credits or forecast capacity with a duration limited by the DSP Dispatch Requirement on business days between 08:00 and 20:00,
    - ESR – assigned Capacity Credits or forecast capacity with duration limited by the duration for which they are assigned or forecast capacity,
    - Intermittent Facilities (wind and solar) – availability traces (which may exceed assigned Capacity Credits), and
    - DER, including those aggregated as VPPs, were not considered on the supply side – instead, their impact was applied to demand.
  - Availability was then limited by Outages, specifically:
    - Planned Outages, being planned maintenance which is provided through FIRs and automatically scheduled by the model for periods without maintenance information, and
    - Forced Outages, randomised outages to align with historical forced Outage rates for that Facility or technology (for committed Facilities).
- Probabilistic assessment:

  - For each modelled Capacity Year, a total of 1,600 simulations were performed to capture a range of operational outcomes arising from:
    - random Forced Outages of thermal generators (100 Monte Carlo iterations), and
    - variability in weather conditions, represented using 16 historical weather reference years to explore weather conditions including demand and wind and solar availability.
- Reliability metric calculation:
  - For each Capacity Year, each of the 1,600 simulation produces an estimate of annual USE. For each Capacity Year, USE was calculated as the average of USE across all simulations.

Approach to determining the Limb B requirement

The Limb B requirement is determined through an iterative modelling process to identify the minimum capacity required to meet the reliability standard (USE $\leq$ 0.0002%) for each Capacity Year.

The steps to determine the Limb B requirement were as follows:

1. Initial modelling – the PLEXOS model was run to dispatch the forecast supply capacity to meet half-hourly demand traces and to calculate USE for each Capacity Year.
2. Assessment against the reliability standard – modelled USE outcomes were compared against the reliability standard and the corresponding Limb A requirement to determine whether further analysis was needed. The criteria for determining whether re-modelling is required are summarised in **Table 13.**

Table 13 Logic of determining the amount of capacity needed to meet the Limb B requirement

Limb B Modelled AIC Compared to Limb A requirement	USE	Re-run required - outcome
Limb B Modelled AIC < Limb A Requirement	Less than 0.0002%	No - Limb A sets the RCT
	More than 0.0002%	Yes - Further modelling required to determine Limb B by adding capacity for reliability (CFR)
Limb B Modelled AIC > Limb A Requirement	Less than 0.0002%	Yes - Further modelling required to determine Limb B by removing Capacity
	More than 0.0002%	Yes - Limb B sets the RCT but further modelling required to determine Limb B

3. Iterative re-modelling (where required).
   - For Capacity Years requiring re-assessment, model runs were repeated with adjusted capacity levels to identify the minimum capacity at which the reliability standard is satisfied. Adjustments were implemented through:
     - reductions in existing capacity (in order of forecast retirement) where surplus capacity is identified, or
     - inclusion of CFR (representing capacity that is always available and not subject to ramping constraints) where reliability shortfalls are observed.
   - This process was repeated until USE was reduced to below the reliability standard. The resulting capacity level represents the Limb B requirement.
4. Determination of the RCT – for each Capacity Year, the derived Limb B requirement was compared with Limb A. The higher of the two values determines the RCT.

A4.3.1 Approach to modelling system constraints

The Limb B dispatch modelling considered a range of system constraints which impact the dispatch of the supply capacity and its ability to meet demand. These constraints include:

- ESS raise constraints, reflective of the need to withhold a portion of capacity for the Regulation Raise and Contingency Reserve Raise services. These constraints effectively limit the ability of supply capacity to meet demand and might result in instances of unserved demand, and thus contribute to annual USE. Contingency Reserve Raise service is dynamic with the state of the system.
- ESS lower constraints, reflective of the need to maintain downward flexibility for the Regulation Lower and Contingency Reserve Lower services. These constraints may result in higher levels of generation or reduced charging to preserve the ability to provide downward response, which can impact dispatch and, to a lesser extent, may contribute to USE due to intertemporal effects (e.g. limit the ability of storage to discharge in peak periods if has not been able to sufficiently charge beforehand). Contingency Reserve Lower service is dynamic with the state of the system.
- ESS Rate of change of Frequency (RoCoF) constraint, reflective of the need to provide inertia services. This constraint requires a minimum number of synchronous Facilities online and is dynamic with the state of the system.
- The Minimum Demand Threshold (MDT) constraint, which requires that operational demand does not fall below 300 MW, as a conservative projected MDT for the outlook period.
- Western Power limit advice 31 and 32, which requires a minimum number of synchronous units to be online at any time in South and the North, respectively.
- DSP constraints, which limits the number of hours in a year that a DSP may operate. Further information on the derivation of this limit is described in Section 4.2 of the 2026 WEM ESOO report.
- Fuel constraints, which limit the daily fuel consumption of gas and distillate fuels. These constraints may be exceeded where USE may occur, and therefore provide an indication of potential fuel limits.

A4.3.2 Approach to modelling network constraints and sub-regional shortfalls

The sub-regional assessment used the same PLEXOS dispatch model as the Limb B assessment described in Appendix A4.3.1, with the addition of network constraint equations and sub-regional generators. This section details the formulation and application of network constraints and sub-regional generators to the Limb B PLEXOS model for the sub-regional assessment. Further analysis on the impact of network constraints on reliability and the impact of network congestion on facility dispatch is described in Section 6 of the 2026 WEM ESOO report.

Power system model

The process to generate network constraint equations for the sub-regional assessment involved running load flow simulations of the SWIS through power system analysis software. A PowerFactory model of the SWIS was obtained from Western Power and used for the constraint formulation process. The model was obtained in March 2026 and represented the most up-to-date power system model of the SWIS.

The SWIS power system model includes the following network configuration:

- Clean Energy Link (CEL) North from the 1 January 2028, and CEL East Enhancement Stage 1 and Stage 2 from 1 October 2026 (start of modelling horizon) and 1 January 2027, respectively. Other CEL Projects, such as the recently announced

priority project CEL East and the soon to be announced priority project CEL Kwinana[39] were not included in the PowerFactory model and not considered in the sub-regional assessment. AEMO expects in future ESOOs that PowerFactory models will include these network augmentation projects and committed generation will utilise these augmentations. For more details on CEL projects refer to Western Power's *Transmission System Plan* (TSP) Appendix B or the Western Australian Government's *SWIS Transmission Plan*[40].

- The model additionally included circuit uprates around the SWIS, such as the Mandurah and Picton load areas. For further details on circuit uprates and other network projects, refer to TSP Appendix B.
- Summer ratings – being the most conservative reflection of network capacity – were used for transmission elements as provided in the PowerFactory model. Winter ratings and dynamic line ratings (DLR) were not included in the sub-regional assessment.
- The PowerFactory model was updated to align with the existing and committed facility list as described in Section 3 of the 2026 WEM ESOO report, Appendix A2 and Appendix A3.

Formulation of network constraints

- Network constraint equations represent linear approximations of power flows across transmission elements due to market dispatchable Facilities. They are formulated as inequalities of the form Ax <= b, and are described further in WEM Procedure: Constraint Formulation[41].
- The transmission elements considered in the sub-regional modelling included all lines with a voltage at or above 66 kilovolts (kV), and all transformers with at least two ends at or above 66 kV. Zone substation transformers were excluded as they are designed to serve downstream loads and are not affected by market dispatch, as is consistent with other forms of WEM network constraint formulation processes. Western Power reports on the utilisation of zone substation transformers in Appendix E of the TSP[42].
- The constraint formulation process applied in the sub-regional assessment is open-loop, similar to ST PASA or MT PASA constraint formulation processes. The process considered the following:
  - the PowerFactory model of the SWIS described above to conduct the power flow simulations,
  - a single system demand term based on scaling the non-block load terms when formulating the constraints,
  - summer ratings for transmission elements as per the provided PowerFactory model,
  - network constraints added to the PLEXOS model in Reverse Polish Notation (RPN) format that included 757 unique network constraints,
  - one additional stability constraint included to limit imports into the Eastern Goldfields, and

[39] See https://www.wa.gov.au/government/media-statements/Cook%20Labor%20Government/Government-plans-WA%27s-clean-energy-future-with-%241.4-billion-fund-20260427.

[40] See https://www.wa.gov.au/organisation/energy-policy-wa/the-south-west-interconnected-system-transmission-plan-future-ready-transmission-network-was-main-electricity-system.

[41] See https://www.aemo.com.au/-/media/files/electricity/wem/procedures/2023/wem-procedure---constraint-formulation---v10---final---external.pdf?la=en&rev=16a65b10aa134f138875d30952698339&sc_lang=en&hash=17D1B9C77555A38963F4892F1E5A6F12.

[42] See https://www.westernpower.com.au/resources-education/suppliers/tenders-and-registrations-of-interest/transmission-system-plan--network-opportunity-map/.

- linearised forms of the existing and future special protection schemes, including runback schemes and the MARNET and 220 kV intertrip schemes.

Sub-regional generators

Sub-regional generators are notional generators added to the PowerFactory and PLEXOS models at sub-regions which may require local generation to relieve network congestion causing USE.

They are modelled as dispatchable sources with infinite capacity and no operational constraints.

These generators are included in the left-hand side of network constraint equations where their contribution to the flow across the respective network is at least 1%.

Sub-regional generators are included in the modelling such that they are dispatched prior to USE but after other generators unless the other generators are behind network or system constraints.

The PLEXOS model also considers an additional sub-regional generator which is located at the regional reference node (RRN). This RRN generator does not contribute to network constraints. The purpose of this generator is to represent generation shortfalls, due to either system-wide supply shortages or network congestion.

Simulations were run in a similar manner as the Limb B runs, with 16 weather reference years and 100 outage scenarios. Dispatch of the RRN sub-regional generator represents the USE as a result of system wide generation shortfalls, including due to network congestion. Dispatch of the other sub-regional generators represents the USE as a result of network congestion in supply to sub-regions of the SWIS, typically due to escalating demand levels resulting in insufficient local generation and transmission capacity. As the network constraints and sub-regional generators were the only changes to the Limb B modelling, these simulations serve as counterfactuals with the Limb B results. The analysis is described in Section 6 of the 2026 WEM ESOO report.

A4.4 Approach to determining the amount of capacity required to satisfy Limb C

The purpose of the assessment against Limb C of the Planning Criterion is to determine the highest forecast Four-Hour Demand Increase (FHDI) and the Flexible RCT.

The highest FHDI quantifies the expected maximum upwards ramp of SWIS unscheduled operational demand over a period of contiguous four hours which needs to be satisfied by an adequate amount of flexible capacity. The assessment of the FHDI uses 10% POE *Expected* and 50% POE *Expected* demand traces.

FHDI is calculated on the basis of half-hourly demand profiles, developed for each Capacity Year based on 16 weather reference years. For a given Capacity Year, the maximum FHDI is calculated as follows:

- For each of the weather reference years, across the entire dataset of 17,520 or 17,568 half-hourly intervals per year, determine individual FHDI as the difference between Trading Interval i and i-8.
- For each of the weather reference years, determine the maximum FHDI.
- The highest forecast FHDI is the average of maximum FHDI for individual weather reference years.

The above steps were implemented using both the 10% POE *Expected* and 50% POE *Expected* demand. The highest forecast FHDI is the greater of the highest forecast FHDI of the two demand scenarios as per clause 4.5.10(bA)(ii) of the ESM Rules.

Similarly, the flexible capacity required to meet Flexible RCT specified in clause 4.5.10(bA)(i) was calculated for both scenarios and the Flexible RCT was determined as the greater of the two. The flexible capacity was determined as the sum of the highest forecast FHDI expected in the relevant Capacity Year and a reserve margin. The reserve margin was calculated in accordance with clause 4.5.9(c) as the highest FHDI multiplied by the proportion of flexible capacity expected to be unavailable at the time of the highest forecast FHDI due to Forced Outages. The Forced Outage rate was estimated using 36 months of historical data for existing Facilities and generic data from the 2025 IASR[43] for new Facilities.

A4.5 Approach to developing Availability Curves

Availability Curves were developed for the second and third Capacity Year, determined by summing the following two items for each interval of a Capacity Year:

a) The forecast demand for the 10% POE *Expected* scenario.

b) A constant margin, applicable to all demand intervals in a Capacity Year, being the difference between the RCT and the forecast annual peak demand for the 10% POE *Expected* scenario.

If RCT was set by Limb A of the Planning Criterion, item (b) of the Availability Curve is equal to the sum of:

- the Intermittent Load allowance,
- the reserve margin,
- the Regulation Raise allowance, and
- the ESR Duration Requirement Uplift.

However, if the RCT were to be set by limb B of the Planning Criterion, item (b) is derived as the difference between:

- the RCT (as set by Limb B), and
- the forecast 10% POE *Expected* annual peak interval demand, that is, the annual maximum value of item (a).

Based on the above, the Availability Curves were developed as follows:

- Ranked demand intervals for the 10% POE *Expected* scenario (average of reference years) in order of descending magnitude of demand.
- Increased each demand data point by adding a constant margin (item (b) above) being the difference between:
  - the value of the RCT, and
  - the value of the forecast 10% POE *Expected* peak demand.

[43] See https://www.aemo.com.au/energy-systems/major-publications/integrated-system-plan-isp/2026-integrated-system-plan-isp/2025-26-inputs-assumptions-and-scenarios.

A4.6 Detailed reliability assessment results

A4.6.1 Determining the Limb B requirement

In line with Step 1 of the Limb B methodology (see Appendix A2.3), initial modelling results for each Capacity Year are presented in **Table 14**.

Table 14 Initial modelling results for Limb B assessment by Capacity Year

Capacity Year	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34	2034-35	2035-36
Limb B Modelled Fleet (MW)	5,987	6,287	6,670	6,310	6,438	5,727	5,588	5,554	5,371	5,293
Limb A requirement (MW)	5,702	6,121	6,323	6,475	6,644	6,807	6,897	6,997	7,197	7,454
Oversupply / shortfall (-) against Limb A	285	166	347	-165	-206	-1,080	-1,309	-1,443	-1,826	-2,161
USE (%)	0.00003	0.00000	0.00000	0.00004	0.00063	0.03041	0.05246	0.08871	0.4037	0.88857
Re-run required	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes

Following application of the re-run process:

- Capacity reductions were applied in early years where surplus capacity was identified. Capacity reductions were applied to coal units in retirement order. Accordingly, in 2026 the capacity of Collie_G1 was reduced by 267 MW, and in 2027 and 2028, Muja_G7 and Muja_G8 respectively were assumed to be retired.
- No re-run was required for 2029-30, as the Limb B modelled fleet fell short of Limb A requirement while meeting the reliability standard.
- From 2030 onwards, additional capacity was introduced via CFR to achieve the reliability standard. CFR is assumed to represent capacity that is always available and not subject to ramping constraints.

Final results are presented in **Table 15**.

Table 15 Final modelled capacity, unserved energy outcomes and determination of binding RCT

Capacity Year	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34	2034-35	2035-36
Limb B Modelled Fleet (MW)	5,987	6,287	6,670	6,310	6,438	5,727	5,588	5,554	5,371	5,293
Limb A requirement (MW)	5,702	6,121	6,323	6,475	6,644	6,807	6,897	6,997	7,197	7,454
Capacity adjustment (MW)	-267	-422	-422	0	193	835	1,032	1,128	1,450	1,815
Post-adjustment USE (%)	0.00019	0.00001	0.00000	0.00004	0.0002[A]	0.00019	0.0002[A]	0.00018	0.0002[A]	0.00018
Final capacity (MW)	5,720	5,865	6,248	6,310	6,631	6,562	6,620	6,682	6,821	7,108
Binding limb	Limb B	Limb A	Limb A	Limb A	Limb A	Limb A	Limb A	Limb A	Limb A	Limb A

A. Values are rounded to five decimal places. Values shown as 0.0002% are marginally below the reliability standard.

B. ESR units are modelled to charge from available system headroom and discharge during periods of unserved energy.

B. ESR units are modelled to charge from available system headroom and discharge during periods of unserved energy.
B. ESR units are modelled to charge from available system headroom and discharge during periods of unserved energy.
C. The equivalent capacity credit is calculated as the sum of ESR capacity and an assumed Relevant Level of renewable capacity, with a Relevant Level of 31% applied to wind and 13% applied to solar.

A4.6.2 Limb B unserved energy characterised by cause

Characterising USE events provides an opportunity to understand the drivers for shortfalls in modelled reliability outcomes. **Figure 25** and **Figure 26** present USE events as bubbles for each contiguous event (scaled for USE MWh), colour-coded according to system conditions at the time, specifically:

- Planned and Forced Outage conditions which exceed the magnitude of the reserve margin component of Limb A (red),
- renewables availability below 15% Capacity Factor during the USE event (blue), and
- Battery SOC for the first USE interval being below 50% (yellow).

The colours where multiple system condition thresholds are met are blended, with green being a combination of low SOC and low renewables output, purple a combination of Outages and low renewables, and brown capturing all three drivers.

Figure 25 shows Limb B USE events from individual modelled iterations in the early years of the outlook period, which sees minimal USE events. This figure highlights larger USE events at higher demand and events without specifically identifiable drivers being present, indicative of overall fleet shortfall at peak demand. The highlighted 850 MWh event in March 2027 is discussed in detail in **Section 5.4** of the report for a single simulation for the events leading to a USE event.

Figure 25 Largest daily unserved energy events from each weather reference year, coloured according to system conditions, 2026-27 to 2029-30

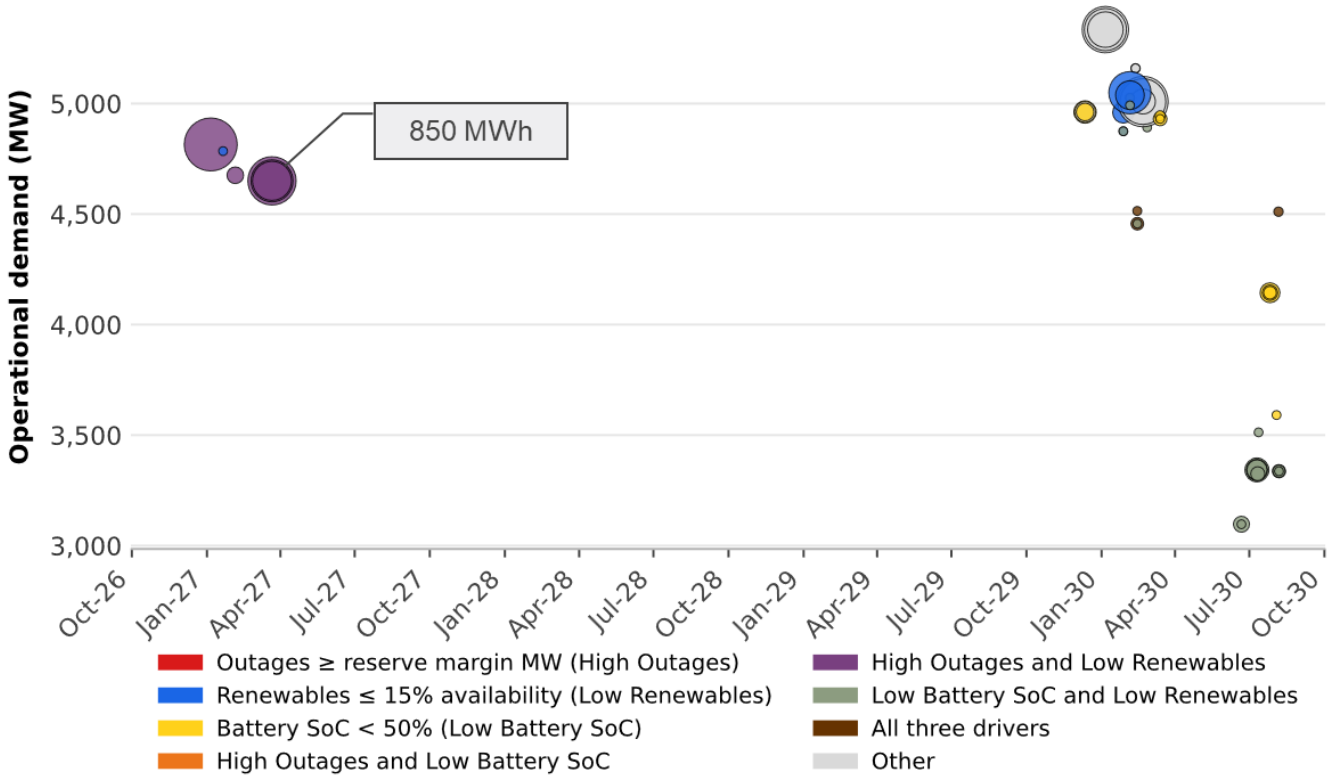
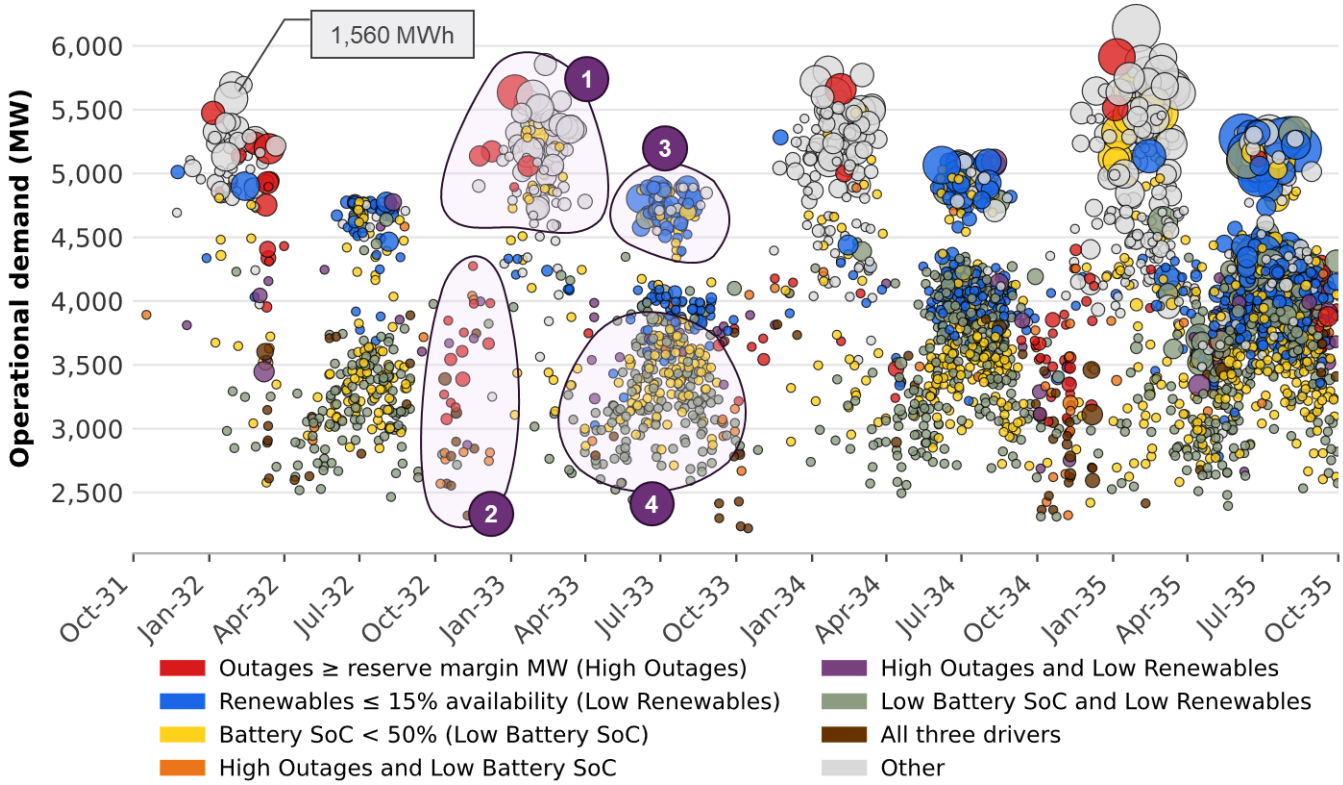


Figure 26 presents outcomes from the outer years of the outlook period. The data shows significantly larger USE outcomes for higher demand levels, and a larger number of events without a specific driver, indicative of overall fleet shortfalls. The larger number of USE events provides clearer patterns of causation, including:

- Summer events generally occur at high demand, occurring without specific causes, indicative of overall shortfall of supply (aligned to supply demand balance for the outer years).
- Shoulder events occur at moderate demand levels due to increased Outages, generally due to higher planned Outages in shoulder seasons.
- Winter events at higher demand levels are dominated by reduced renewables availability.
- Events in winter also occur at low demand, where low renewables combined with low SOC result in USE even at low demand.

Figure 26 Largest daily unserved energy events from each weather reference year, coloured according to system conditions, 2031-32 to 2034-35



A4.6.3 Unserved energy by weather reference year and by month

Unserved energy by weather reference years

Table 16 provides a breakdown of USE by weather reference year for each Capacity Year in the outlook period.

The weather patterns associated with 2011-12, 2012-13, 2016-17 and 2017-18 weather reference years result in higher USE than other weather years modelled.

After 2030-31, the general shortfall among all historical weather reference years suggests that USE is increasingly driven by a combination of increasing consumption, retirements, and no new entrant capacity.

The impact of a weather reference year pattern on USE outcomes for a given Capacity Year is largely coincident with planned maintenance of thermal plant. In events where planned maintenance occurs and the given weather reference year pattern results in low renewable generation during the maintenance period, the likelihood of USE increases.

For the 2030-31 Capacity Year, two facilities have overlapping planned maintenance during February. For this Capacity Year, the 2016-17 weather reference year pattern coincides with low renewable generation in February, resulting in a high amount of USE occurring for this Capacity Year and the 2016-17 weather reference year.

Table 16 Annual unserved energy (MWh) by weather reference year for each Capacity Year

Weather reference year/ Capacity Year	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34	2034-35	2035-36
2009-10	0	0	0	0	1	4,808	8,138	22,174	122,410	229,553
2010-11	0	0	0	0	161	11,109	17,602	21,901	112,328	221,121
2011-12	0	0	0	15	100	15,618	20,647	33,365	85,712	221,062
2012-13	0	0	0	41	38	8,705	17,703	21,027	137,902	311,716
2013-14	0	0	0	0	48	2,004	3,090	5,510	34,489	163,379
2014-15	10	0	0	41	17	4,346	12,481	19,882	95,755	220,694
2015-16	0	0	0	0	0	2,679	5,773	12,517	61,876	229,489
2016-17	0	0	0	1	1,596	9,336	10,714	23,482	144,618	271,596
2017-18	73	0	0	4	17	14,628	20,838	30,444	158,631	246,020
2018-19	0	0	0	1	1	3,343	8,409	26,682	142,511	313,250
2019-20	0	0	0	0	7	2,860	6,829	24,628	96,127	175,789
2020-21	0	0	0	0	0	6,286	10,911	15,501	63,410	222,174
2021-22	0	0	0	0	6	2,215	4,856	6,339	46,285	154,035
2022-23	1	0	0	10	29	4,520	9,069	12,847	85,732	211,068
2023-24	0	0	0	0	41	3,522	12,908	20,165	82,006	184,150
2024-25	0	0	0	12	1	6,860	13,042	25,185	75,432	234,441
Total USE	84	0	0	125	2,065	102,840	183,010	321,648	1,545,223	3,609,538

Unserved energy by month

Table 17 presents USE by month for each Capacity Year over the 2026 Long Term PASA Study Horizon.

For 2026-27, the results show that USE occurs only during summer months, which typically have the highest demand.

From 2029-30, the occurrence of USE is seen in both summer and winter months. From 2031-32, USE begins to emerge in shoulder months. From 2032-33 USE occurs in all months of the year from 2032-33. From 2033-34 onwards, USE becomes most dominant in winter, due to a lack of sufficient supply and lower renewable penetration.

Table 17 Unserved energy (MWh) by month and Capacity Year

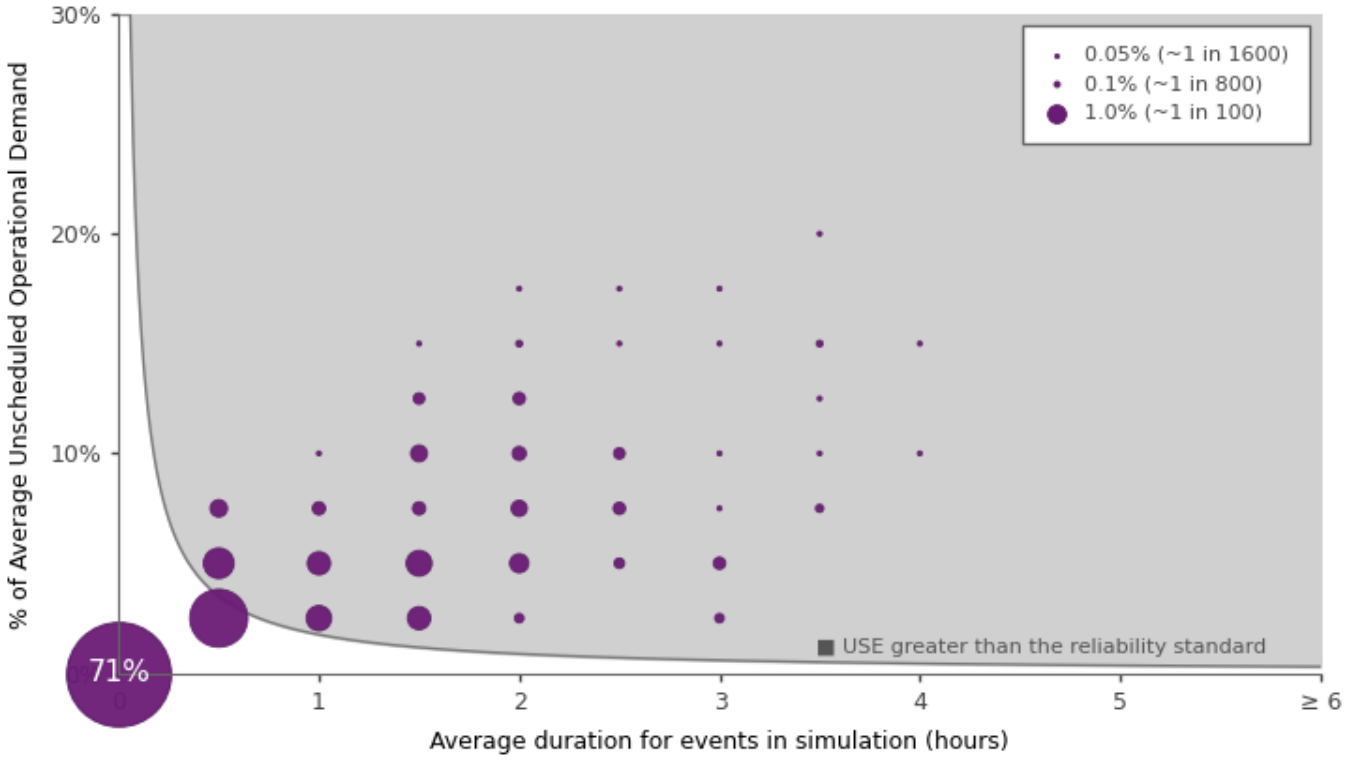
Month/ Capacity Year	2026-27	2027-28	2028-29	2029-30	2030-31	2031-32	2032-33	2033-34	2034-35	2035-36
October	0	0	0	0	0	0	37	4	79	11,701
November	0	0	0	0	3	68	260	201	564	4,780
December	0	0	0	1	0	81	200	289	1,095	3,607
January	1	0	0	3	16	1,702	2,982	3,689	10,453	22,776
February	0	0	0	2	98	1,560	2,181	3,351	10,970	19,019
March	5	0	0	0	3	1,597	1,530	1,997	5,653	10,314
April	0	0	0	0	0	3	13	76	1,276	3,851
May	0	0	0	0	0	47	66	116	1,888	11,641
June	0	0	0	0	0	520	1,533	4,328	23,286	52,179
July	0	0	0	2	2	332	1,300	3,352	28,439	58,562
August	0	0	0	0	6	517	1,306	2,644	12,029	24,245
September	0	0	0	0	0	1	28	57	843	2,920
Total USE	**6**	**0**	**0**	**8**	**128**	**6,428**	**11,436**	**20,104**	**96,575**	**225,595**

A4.6.4 Unserved energy events by duration, magnitude and frequency

Figure 27 shows a bubble plot of the distribution of USE outcomes that are forecast for the 2030-31 Capacity Year. The figure plots the USE duration and average magnitude represented as a percentage of unscheduled operational demand of each USE event of the 1,600 simulations. The area of each bubble represents the probability of a USE event occurring with the given duration and magnitude. The grey section presents a boundary within which any duration and magnitude would exceed the reliability standard of 0.0002% USE. Bubbles that occur inside of the grey area exceed the 0.0002% reliability standard. The figure shows the following:

- The most likely outcome for the 2030-31 Capacity Year is that no USE events will occur, with this being the case in 71% of simulations.
- Should USE occur in the simulations, it is most likely to be forecast for between half an hour and 2 hours and be of an average USE magnitude equivalent to between 0% and 10% of average Unscheduled Operational Demand.
- There is a very low probability for USE to be forecast as deep as 20% of average demand, or as long as four hours.

Figure 27 Simulated unserved energy duration, magnitude and frequency for 2030-31



Note: Zero-point bubble is not to scale.

A4.6.5 Extended analysis of unserved energy events

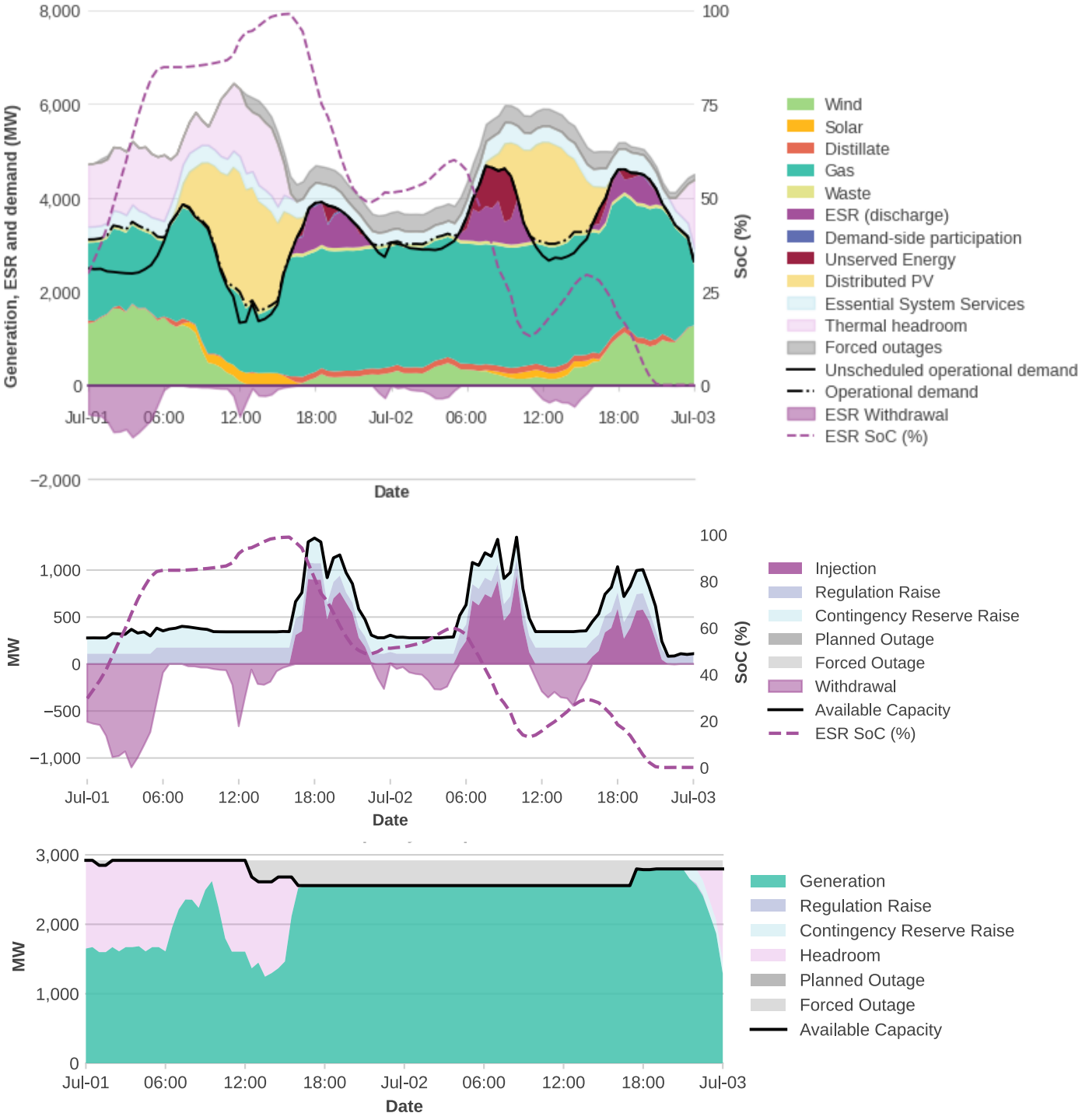
Figure 28 provides a deep dive of a two-day period outlined in Section 5.4 of the 2026 WEM ESOO, where USE events are modelled across multiple days in 2030-31 (sample days in July 2032, reference year 2009-10). The figure includes a more detailed dispatch level view, including breakdown of ESS and technology types.

Interval level dispatch is presented alongside time-aligned data for batteries and gas-fired generation, demonstrating the utilisation of these technologies across the period. The figure highlights the utilisation of storage across ESS and energy, with the model seeking to minimise the magnitude (MWh) of a USE event, which results in more modest but longer duration USE events occurring across multiple days.

The gas generation utilisation highlights headroom on the gas fleet ahead of USE events, and full utilisation during and between the USE events, as the modelling seeks to charge batteries.

The need for sustained operation of gas further highlights the need for detailed assessments of adequacy of both contracted and physical gas supply and infrastructure, given the critical role gas will play during system stress periods.

Figure 28 Modelled outcomes across two-day period capturing an unserved energy event (top), battery utilisation (middle), and gas utilisation (bottom)



Glossary, measures, and abbreviations

Glossary

This document uses many terms that have meanings defined in the Electricity System and Market Rules (ESM Rules). The ESM meanings are adopted unless otherwise specified.

Term	Definition
10-year outlook period	2026-27 to 2035-36 Capacity Years, inclusive.
Business mass market (BMM)	BMM covers those business loads that are not included in the large industrial load sector.
Business sector	Business sector includes industrial and commercial users. This sector is subcategorised further to include large industrial loads, data centres, new industries and business mass market (BMM).
Committed capacity	Capacity includes new projects that are candidates for registration and have been assigned Capacity Credits for 2026-27 or have scored 80% or higher in the new project status evaluation. This category also includes Facilities contracted for the 2025-26 Peak Demand NCESS, and Facilities contracted or expected to be contracted for 2025-27 Peak Demand NCESS.
Consumption	The amount of power used over a period of time, conventionally reported as megawatt hours (MWh), gigawatt hours (GWh), or terawatt hours (TWh), depending on the magnitude of power consumed. It is reported on a "sent-out" basis (excluding electricity used by a generator[44]) unless otherwise stated.
Delivered consumption (or demand)	The total amount of electricity supplied to customers from the grid, which excludes the portion of their consumption/demand that is met by behind-the-meter (typically distributed PV) generation.
Demand	The amount of power consumed at any time. Peak and minimum demand is measured in megawatts (MW) and averaged over a 30-minute period. It is reported on a "sent-out" basis (excluding electricity used by a generator) unless otherwise stated.
Distributed energy resources (DER)	Smaller-scale, geographically dispersed, customer-owned electrical devices that generate, store or use electricity, and form a part of the local electricity distribution system. Examples include rooftop photovoltaics (PV) systems, battery storage systems (residential, business, community), electric vehicles (EV) or technologies to manage demand at a premises. DER are located "behind-the-meter", and used within residential or business premises to self-supply co-located loads such as air conditioners, heat pumps, or washing machines. If on-site supply exceeds on-site load, DER may export (inject) power to the broader system. If coordinated through a virtual power plant (VPP) arrangement, DER can support the operations of the broader power system. Subject to meeting criteria for Capacity Credits assignment, DER may hold Capacity Credits in the WEM.
Distributed energy storage systems (DESS), or distributed batteries	A sub-group of DER, DESS are behind-the-meter, smaller-scale battery energy storage systems installed by residential or business customers.
Distributed photovoltaics (PV)	A sub-group of DER, distributed PV are behind-the-meter, smaller-scale PV systems, including rooftop PV and PV non-scheduled generation (PVNSG).
Electric vehicle (EV)	A sub-group of DER, EV are vehicles with an on-board electricity storage system (battery) used for propulsion. EV include a broad range of vehicles, from motorbikes and passenger cars to commercial trucks and buses.
Electrification	Replacement of technologies or processes that use fossil fuels, like internal combustion engines and gas boilers, with electrically-powered equivalents, such as electric vehicles or heat pumps.

[44] This may be called 'auxiliary load', 'parasitic load', or 'self-load', and refers to energy generated for use within power stations.

Term	Definition
Operational consumption[45] (or demand) [46]	Electricity consumption or demand met by sent-out electricity injected into the network by all market-registered energy producing systems. Operational consumption or demand includes the charging and discharging of distributed batteries (which increases and decreases demand, respectively), as well as transmission and distribution losses. Operational consumption or demand does not include consumption or demand met by distributed PV generation.
Unscheduled operational consumption (or demand)[47]	Operational consumption or demand excluding scheduled loads, such as large-scale ESR charging.
Existing capacity	Capacity provided by Registered Facilities that have been assigned Capacity Credits for 2025-26 or 2026-27 and reflecting any announced or anticipated retirements. Existing capacity is included in the low, expected, and high scenarios for the capacity supply forecasts.
Installed capacity	The generating capacity (in MW) of a single or multiple generating units.
Large industrial load	Users that consume, or are forecast to consume, at least 10 MW for a minimum of 10% of the time (around 875 hours a year) or at least 50 GWh per year. large industrial loads include existing and new large industrial loads.
Limb A	Term attributed to the requirement of the Planning Criterion that stipulates that there should be sufficient available capacity in each Capacity Year to meet the forecast peak demand plus a reserve margin.
Limb B	Term attributed to the requirement of the Planning Criterion that stipulates there should be sufficient available capacity in the SWIS to limit expected unserved energy (USE) shortfalls to 0.0002% of annual energy consumption.
Limb C	Term attributed to the requirement of the Planning Criterion that stipulates that there should be enough capacity available in the SWIS to meet the highest forecast Four-Hour Demand Increase, plus a reserve margin, in each Capacity Year.
Maximum capacity	Refers to the maximum dispatchable capacity of a Facility to meet Limb B of Planning Criterion, either stemming from Facilities' nameplate capacity or the contracted quantity through 2025-26 and 2025-27 Peak Demand NCESS or forecast peak capacity. The net sent-out generation or installed capacity of a facility, as detailed on AEMO's Market Data website.
Operational maximum (peak) and minimum demand	The highest and lowest level of electricity drawn from the grid, measured as an average over a 30-minute period in either summer (December to March[48]), winter (June to August), or shoulder months (April, May, September to November).
Peak demand	The highest amount of demand consumed at any one time. Peak demand refers to operational peak demand unless otherwise stated.
Photovoltaics	Systems to convert sunlight into electricity.
Probability of exceedance (POE)	A measure of the likelihood of a value being met or exceeded. For example, a 10% POE maximum demand forecast is expected to be met or exceeded, on average, one year in 10, while a 90% POE maximum demand forecast is expected to be met or exceeded nine years in 10.
Probable capacity	Capacity comprised of new projects that: • are a candidate for registration and have submitted a valid Expression of Interest for the 2026 Reserve Capacity Cycle (2026 EOI)[49], and • have scored 50% or more but less than 80% in the new project status evaluation. Probable capacity is included only in the *High* scenario for the capacity supply forecasts.
Proposed capacity	Capacity includes all new projects that have been proposed but have not met the criteria to be in the existing, committed, or probable capacity categories.

[45] Historical operational consumption is measured as the Total Sent Out Generation (TSOG) over a 30-minute Trading Interval. It is a non-network-loss adjusted MWh value.

[46] Historical operational demand is calculated as the TSOG multiplied by two, to convert MWh to MW for a 30-minute Trading Interval. The historical operational peak demand and minimum demand are identified as the highest and lowest operational demand calculated for a Trading Interval in a Capacity Year, respectively.

[47] Unscheduled operational consumption/demand terms are also defined in the undertaking of the Long Term PASA WEM Procedure, available at https://www.aemo.com.au/energy-systems/electricity/wholesale-electricity-market-wem/procedures-policies-and-guides/procedures.

[48] These months are aligned with the Hot Season defined in the ESM Rules.

[49] The information that a 2026 EOI must include to be deemed valid is outlined in clause 4.4.1 of the ESM Rules.

Term	Definition
PV non-scheduled generation (PVNSG)	Non-scheduled photovoltaic generators larger than 100 kilowatts (kW) but smaller than 10 megawatts (MW) that do not hold Capacity Credits in the WEM.
Reference year	Future half-hourly demand, wind and solar PV generation is modelled based on several historical reference years to capture a variety of Australian weather patterns.
Reliability Standard	The Planning Criterion defined in clause 4.5.9 of the ESM Rules.
Residential sector	Includes non-contestable[50] residential customers (supplied by Synergy) only.
Rooftop photovoltaics	Photovoltaics installed on a residential building (less than 15 kW) or business premises (less than 100 kW).
Shoulder season	The period covering the Trading Months of April, May, September, October and November.
Summer	The Hot Season as defined in the ESM Rules, covering the Trading Months of December, January, February, and March.
Underlying consumption (or demand)	The total amount of electricity consumption (or demand) used by consumers at their power points. This electricity can be sourced from the grid, or from behind-the-meter distributed energy resources (DER) such as distributed photovoltaics (distributed PV) and battery storage.
Unscheduled operational maximum (peak) and minimum demand	This represents the operational peak and minimum demand forecasts that exclude the impact of scheduled load operations (such as ESR charging).
Unserved energy (USE)	Energy that cannot be supplied to consumers when demand exceeds supply under certain circumstances for loss of customer supply.
Virtual power plant (VPP)	An aggregation (grouping) of DER that is actively controlled by an operator to coordinate their behaviour with consideration for, and in response to, the broader power system and electricity market conditions. VPPs use sophisticated software and communication platforms that coordinate and control DER to function as a single, unified facility. This enables the VPP to provide system and network services, participate in wholesale energy markets, and respond to real-time energy demand and supply fluctuations.
Winter	The period covering the Trading Months of June, July and August.

Units of measure

Abbreviation	Unit of measure
°C	Celsius
GW	Gigawatt/s
GWh	Gigawatt hour/s
kV	Kilovolt/s
kW	Kilowatt/s
MW	Megawatt/s
MWh	Megawatt hour/s
TW	Terawatt/s
TWh	Terawatt hour/s

[50] A non-contestable customer is a customer that uses less than 50 MWh of electricity per year and is connected to Western Power's distribution network.

Abbreviations

Abbreviation	Definitions
ADG	Availability Duration Gap
AEMO	Australian Energy Market Operator
BMM	Business mass market
CBD	Commercial Building Disclosure
CDD	Cooling Degree Day
CEL	Clean Energy Link
CFR	Capacity for Reliability
CHBP	Cheaper Home Batteries Program
CRC	Certified Reserve Capacity
CSIRO	Commonwealth Scientific and Industrial Research Organisation
DAE	Deloitte Access Economics
DER	Distributed energy resources
DESS	Distributed energy storage systems
DSP	Demand Side Programme
EOI	Expression of Interest
EPA	Environmental Protection Authority
ESM	Electricity System and Market
ESOO	Electricity Statement of Opportunities
ESR	Electric Storage Resource
ESRDR	Electric Storage Resource Duration Requirement
ESROI	Energy Storage Resource Obligation Interval
ESS	Essential System Services
EV/s	Electric vehicle/s
FCESS	Frequency Control Essential System Services
FHDI	Four-Hour Demand Increase
FID	Final Investment Decision
FIR	Formal Information Request
FRG	Forecasting Reference Group
GEMS	Greenhouse and Energy Minimum Standards
IASR	Inputs, Assumptions and Scenarios Report
IGS	Intermittent Generating Systems
IRCR	Individual Reserve Capacity Requirement
MDT	Minimum Demand Threshold
NABERS	National Australian Built Environment Rating System
NAQ	Network Access Quantity
NCC	National Construction Code
NCESS	Non-Co-optimised Essential System Service
NEM	National Electricity Market

Abbreviation	Definitions
NIGS	Non-Intermittent Generating Systems
NMI	National Metering Identifier
NSS	Network Support Services
NVES	New Vehicle Efficiency Standard
OCGT	Open Cycle Gas Turbine
OPSO	Operational sent out
PASA	Projected Assessment of System Adequacy
PHEV	Plug-in Hybrid Electric Vehicle
POE	Probability of exceedance
PV	Photovoltaic
PSSR	Power System Security & Reliability
PV	Photovoltaic
PVNSG	Photovoltaic non-scheduled generator
RCT	Reserve Capacity Target
RLM	Relevant Level Method
RoCoF	Rate of Change of Frequency
RPN	Reverse Polish Notation
RRN	Regional Reference Node
SOC	State of Charge
SRES	Small-scale Renewable Energy Scheme
STC	Small-scale technology certificates
SWIS	South West Interconnected System
TSOG	Total Sent Out Generation
TSP	Transmission System Plan
TNI	Transmission Node Identifier
USE	Unserved Energy
VPP	Virtual power plant
VRE	Variable renewable energy
WEM	Wholesale Electricity Market